Nebico Oil Retention Project (4.0)

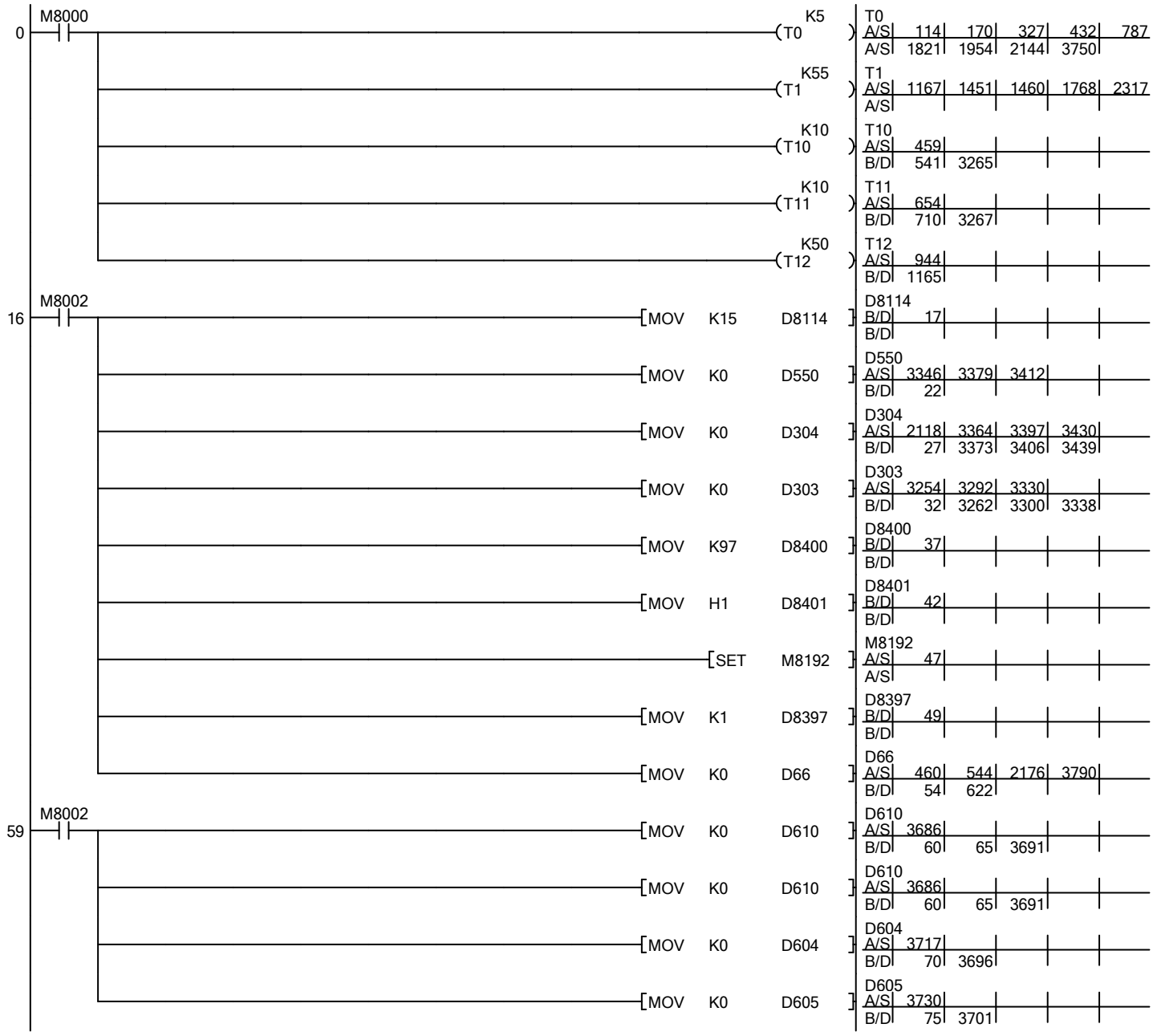
Coolmay L10S Ladder program
TK8050 x 3, TK8100 x 1

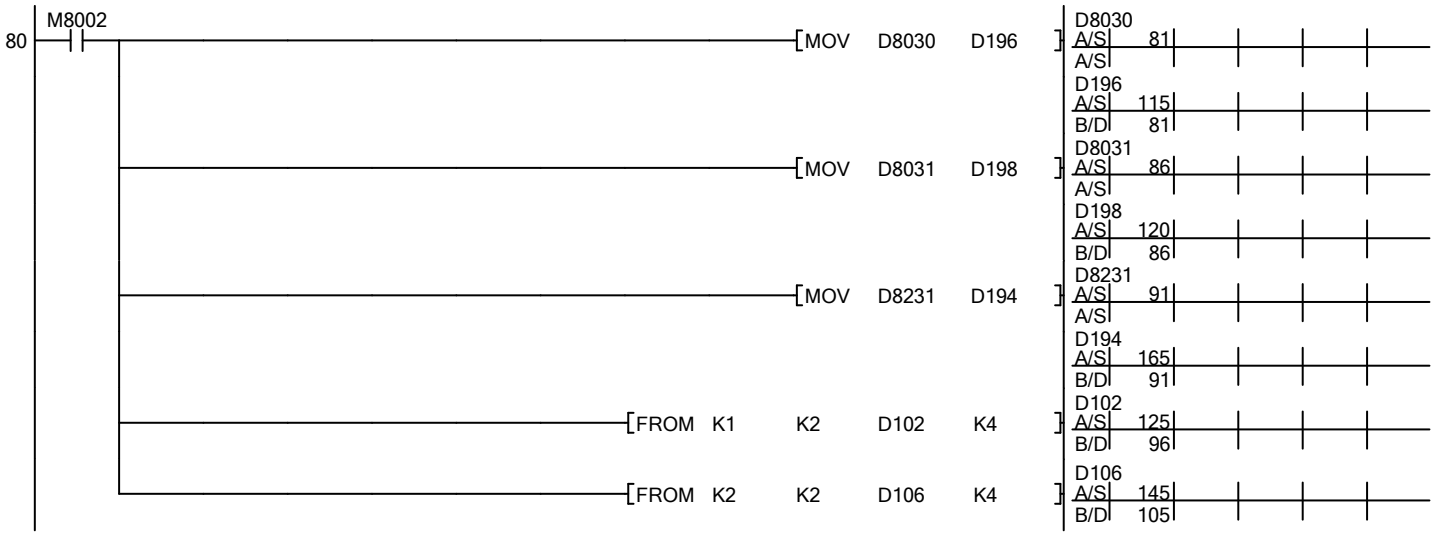
Modification:

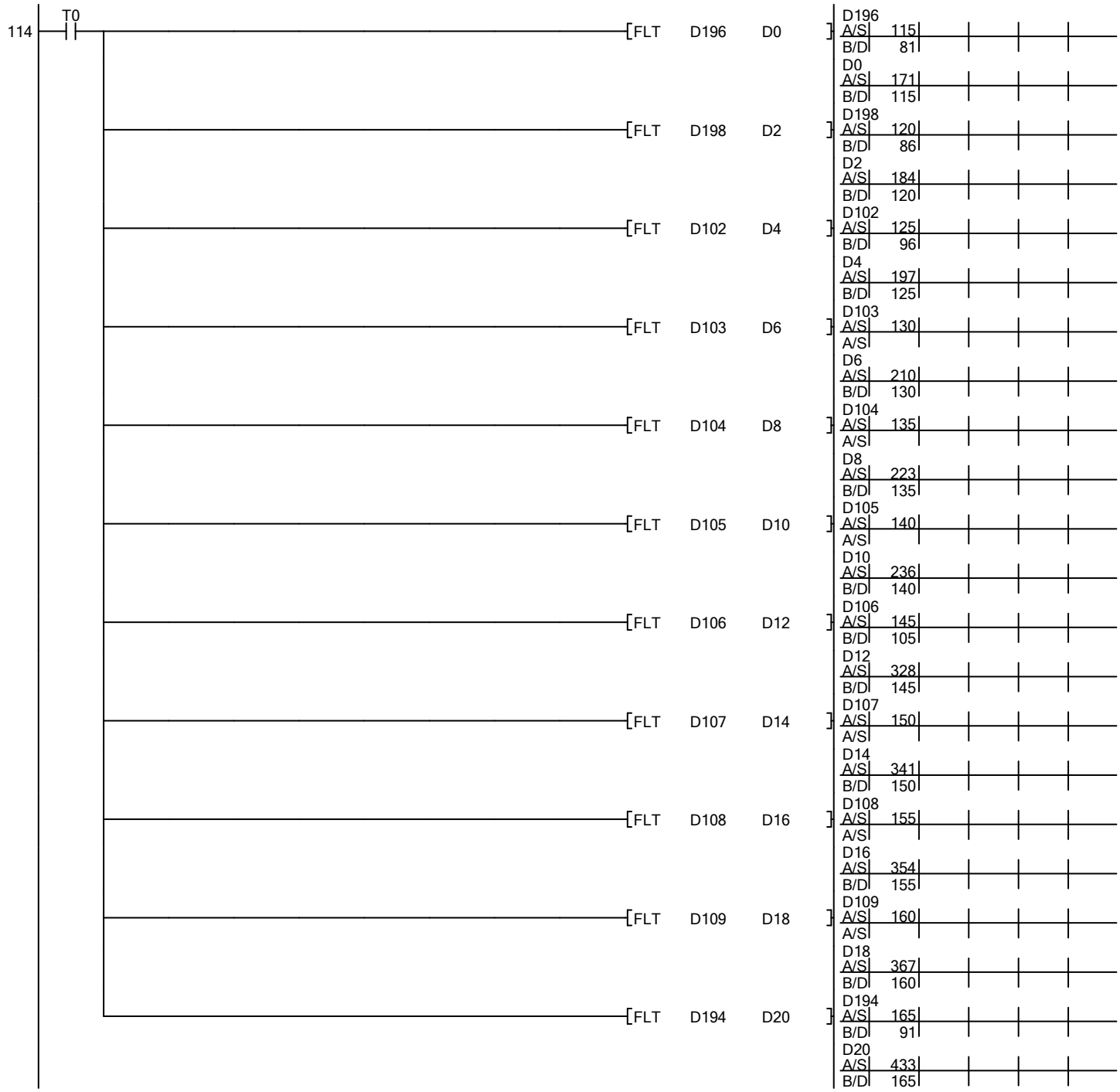
1. Removal of Z-Pointers
2. Max utilisation of General purpose data register

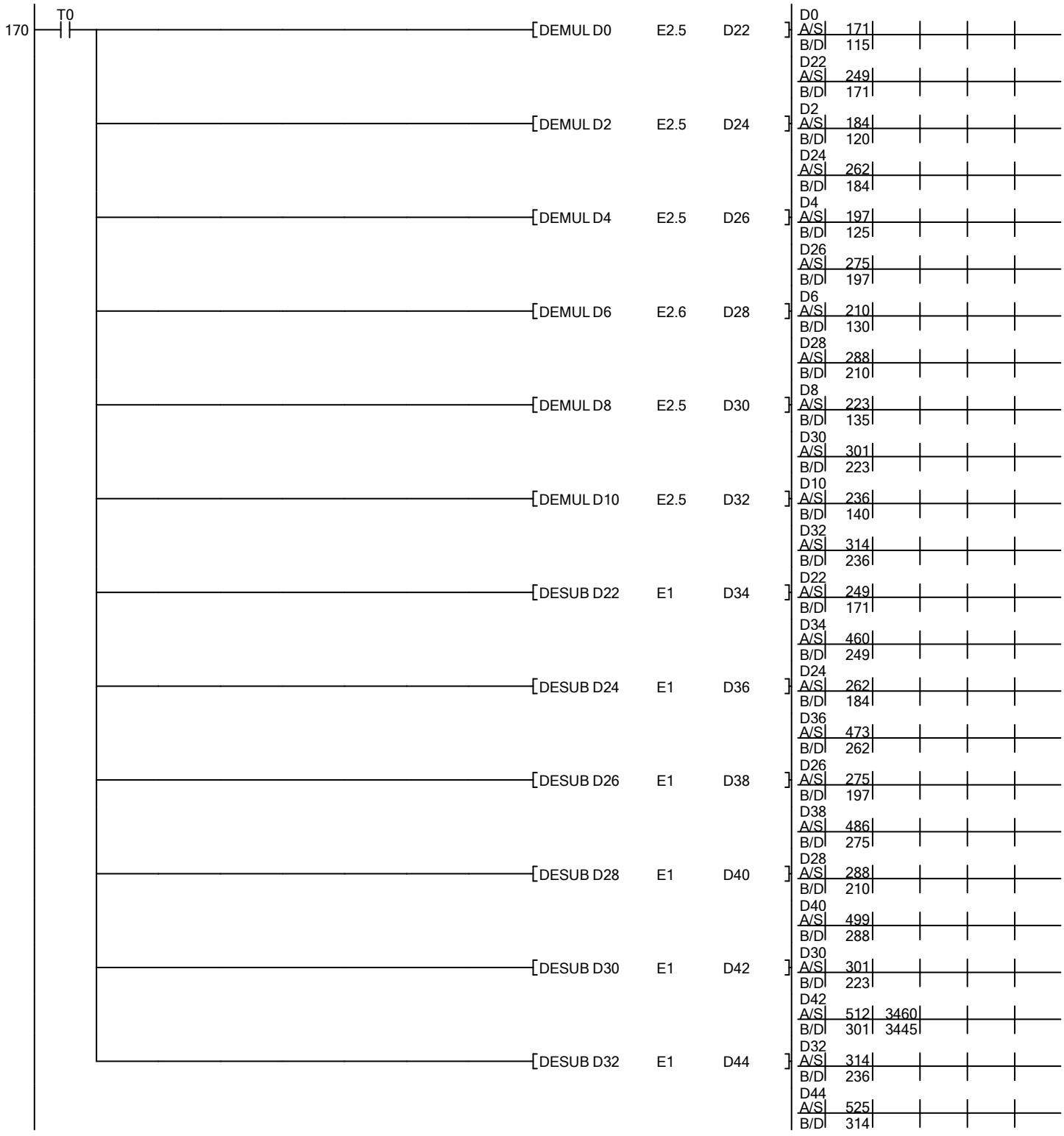
Prepared By: Sambridda Ranabhat (Architect & Programmer)

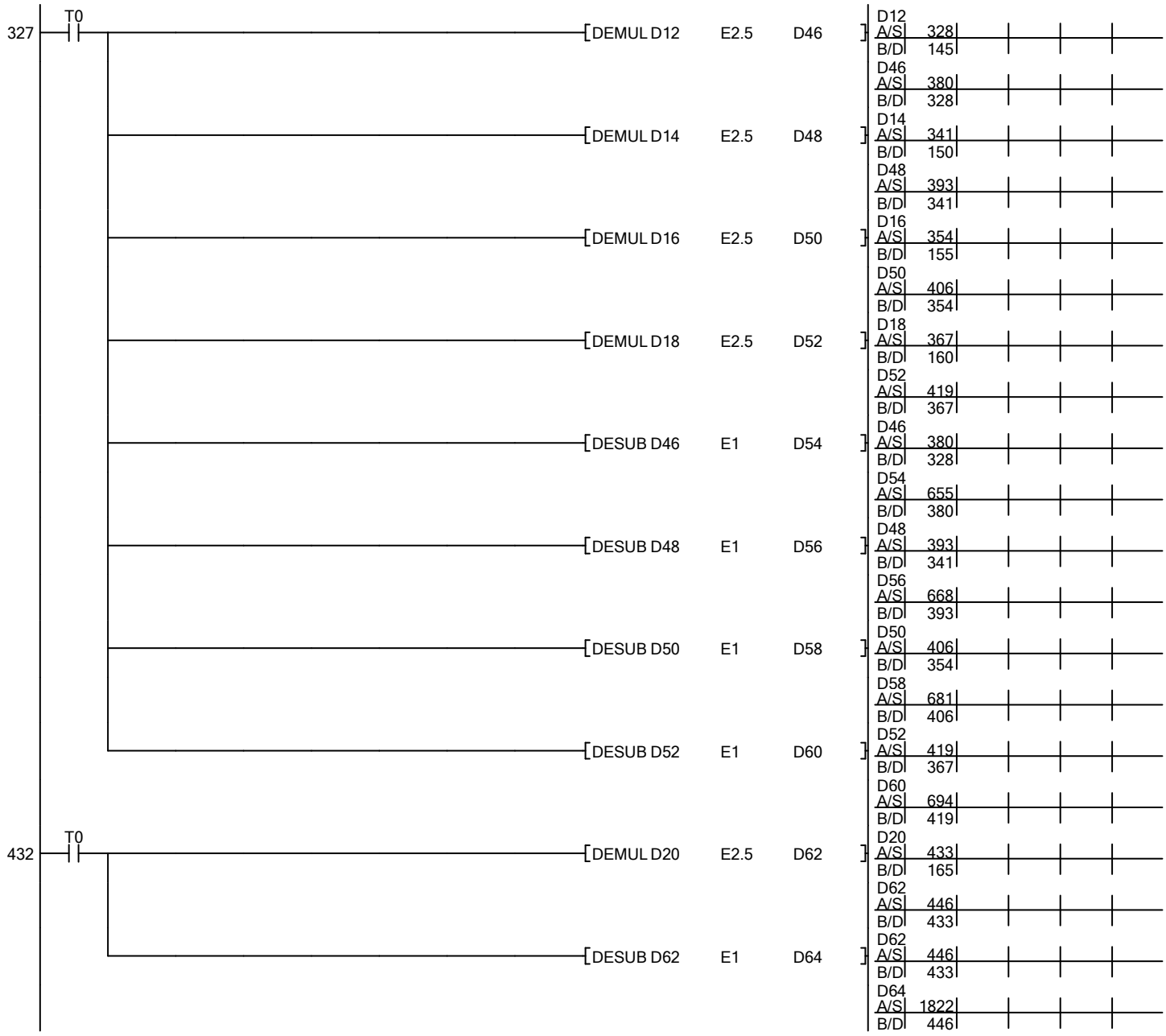
20/09/2026
Mechatronics Engineering .pvt .ltd

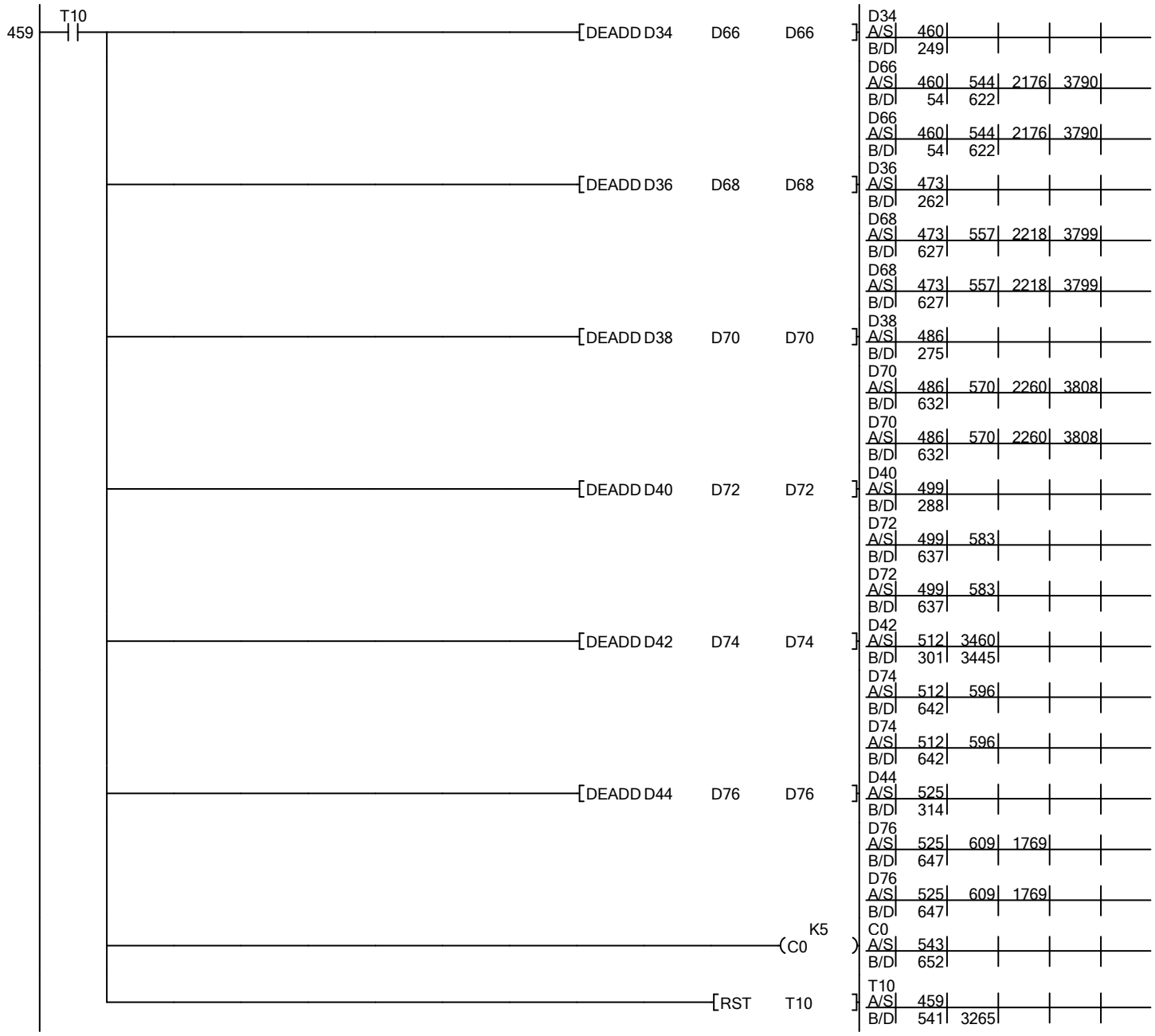

0
M8000
K5
T0
T0
A/S 114 170 327 432 787
A/S 1821 1954 2144 3750
K55
T1
T1
A/S 1167 1451 1460 1768 2317
A/S
K10
T10
T10
A/S 459
B/D 541 3265
K10
T11
T11
A/S 654
B/D 710 3267
K50
T12
T12
A/S 944
B/D 1165
16
M8002
MOV K15 D8114
D8114
B/D 17
B/D
MOV K0 D550
D550
A/S 3346 3379 3412
B/D 22
MOV K0 D304
D304
A/S 2118 3364 3397 3430
B/D 27 3373 3406 3439
MOV K0 D303
D303
A/S 3254 3292 3330
B/D 32 3262 3300 3338
MOV K97 D8400
D8400
B/D 37
B/D
MOV H1 D8401
D8401
B/D 42
B/D
SET M8192
M8192
A/S 47
A/S
MOV K1 D8397
D8397
B/D 49
B/D
MOV K0 D66
D66
A/S 460 544 2176 3790
B/D 54 622
59
M8002
MOV K0 D610
D610
A/S 3686
B/D 60 65 3691
MOV K0 D610
D610
A/S 3686
B/D 60 65 3691
MOV K0 D604
D604
A/S 3717
B/D 70 3696
MOV K0 D605
D605
A/S 3730
B/D 75 3701

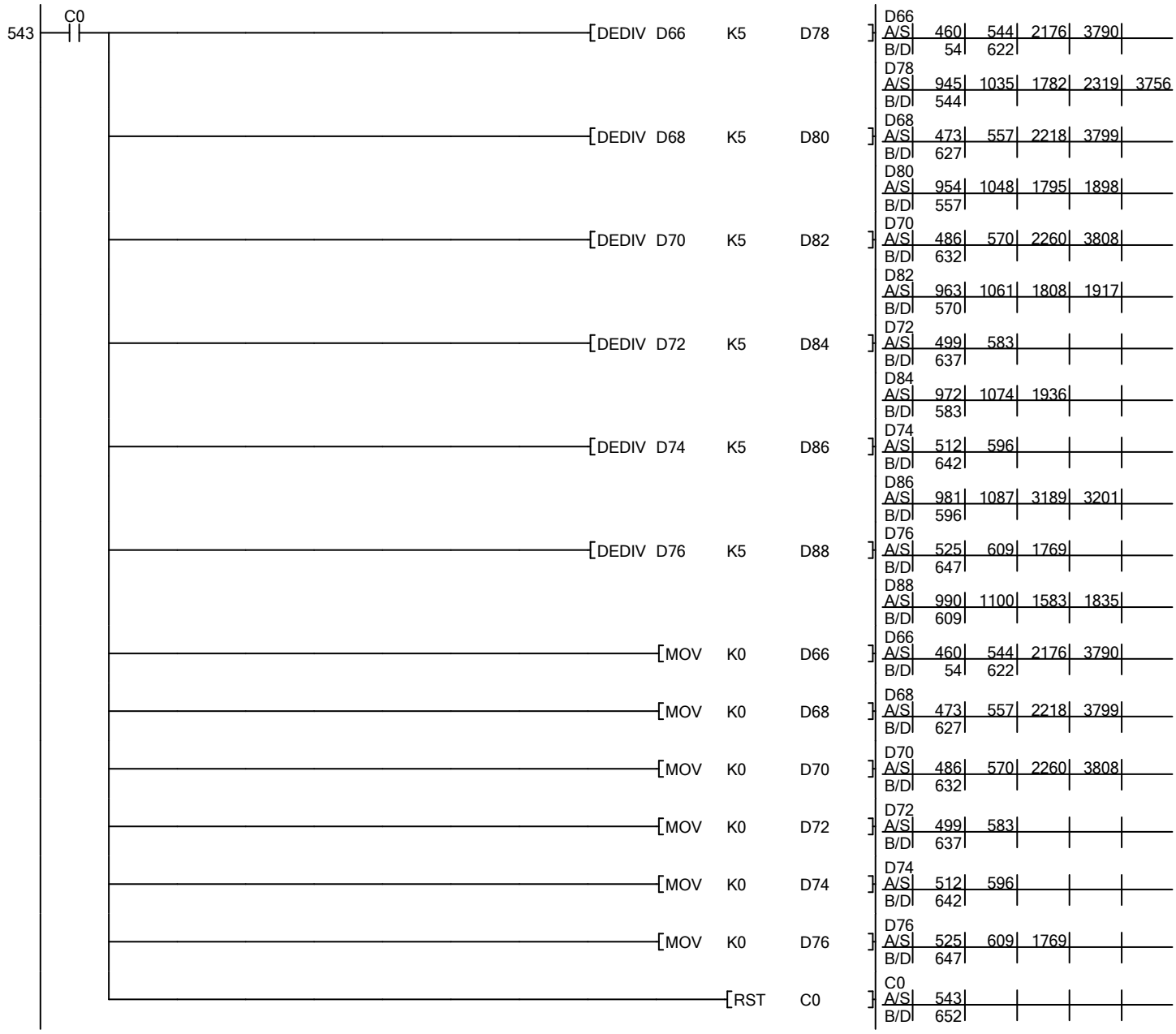

80
M8002
MOV D8030 D196
MOV D8031 D198
MOV D8231 D194
FROM K1 K2 D102 K4
FROM K2 K2 D106 K4
D8030
A/S 81
A/S
D196
A/S 115
B/D 81
D8031
A/S 86
A/S
D198
A/S 120
B/D 86
D8231
A/S 91
A/S
D194
A/S 165
B/D 91
D102
A/S 125
B/D 96
D106
A/S 145
B/D 105

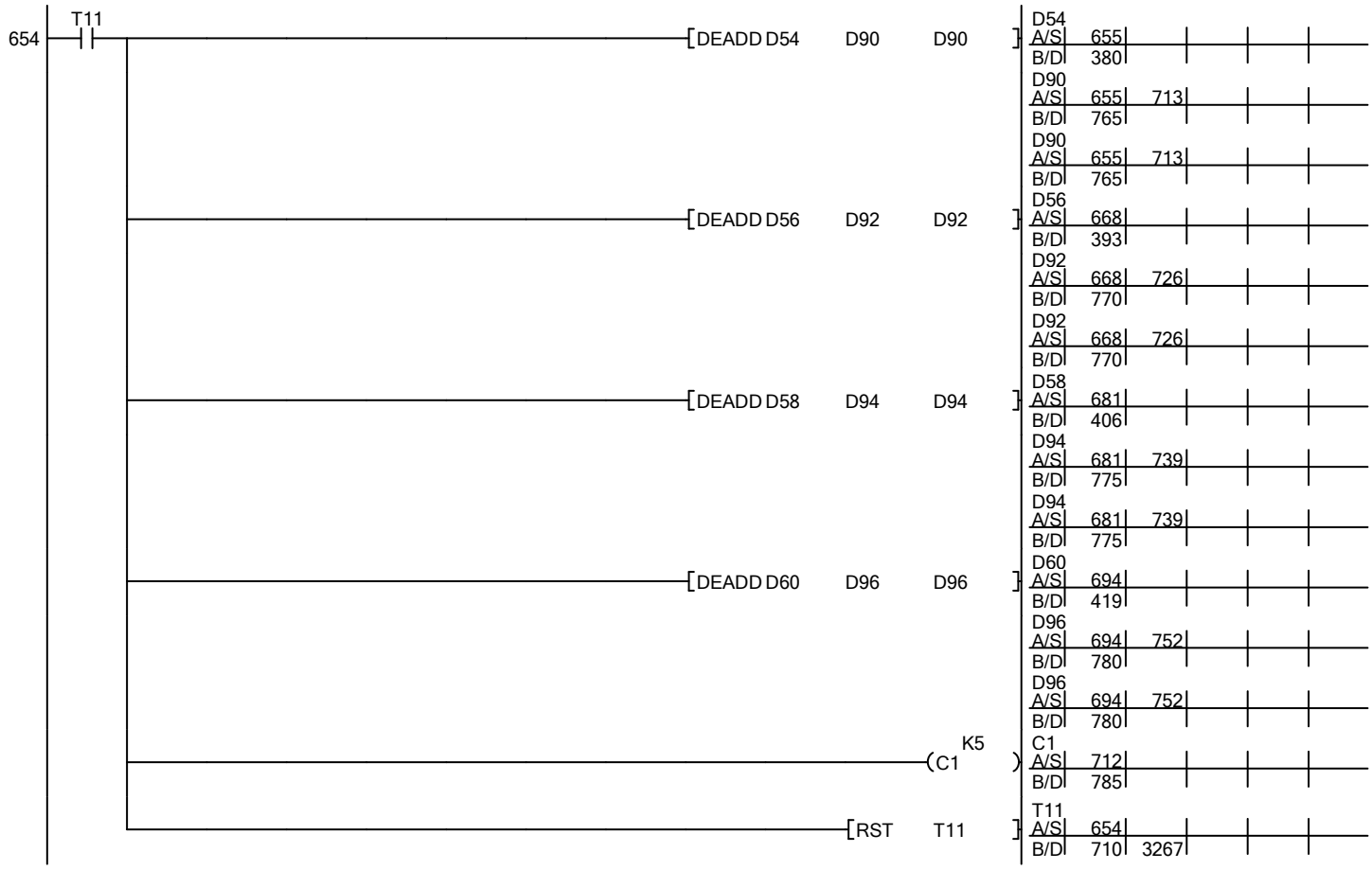

114
T0
FLT D196 D0
FLT D198 D2
FLT D102 D4
FLT D103 D6
FLT D104 D8
FLT D105 D10
FLT D106 D12
FLT D107 D14
FLT D108 D16
FLT D109 D18
FLT D194 D20
D196 A/S 115 B/D 81
D0 A/S 171 B/D 115
D198 A/S 120 B/D 86
D2 A/S 184 B/D 120
D102 A/S 125 B/D 96
D4 A/S 197 B/D 125
D103 A/S 130 A/S
D6 A/S 210 B/D 130
D104 A/S 135 A/S
D8 A/S 223 B/D 135
D105 A/S 140 A/S
D10 A/S 236 B/D 140
D106 A/S 145 B/D 105
D12 A/S 328 B/D 145
D107 A/S 150 A/S
D14 A/S 341 B/D 150
D108 A/S 155 A/S
D16 A/S 354 B/D 155
D109 A/S 160 A/S
D18 A/S 367 B/D 160
D194 A/S 165 B/D 91
D20 A/S 433 B/D 165

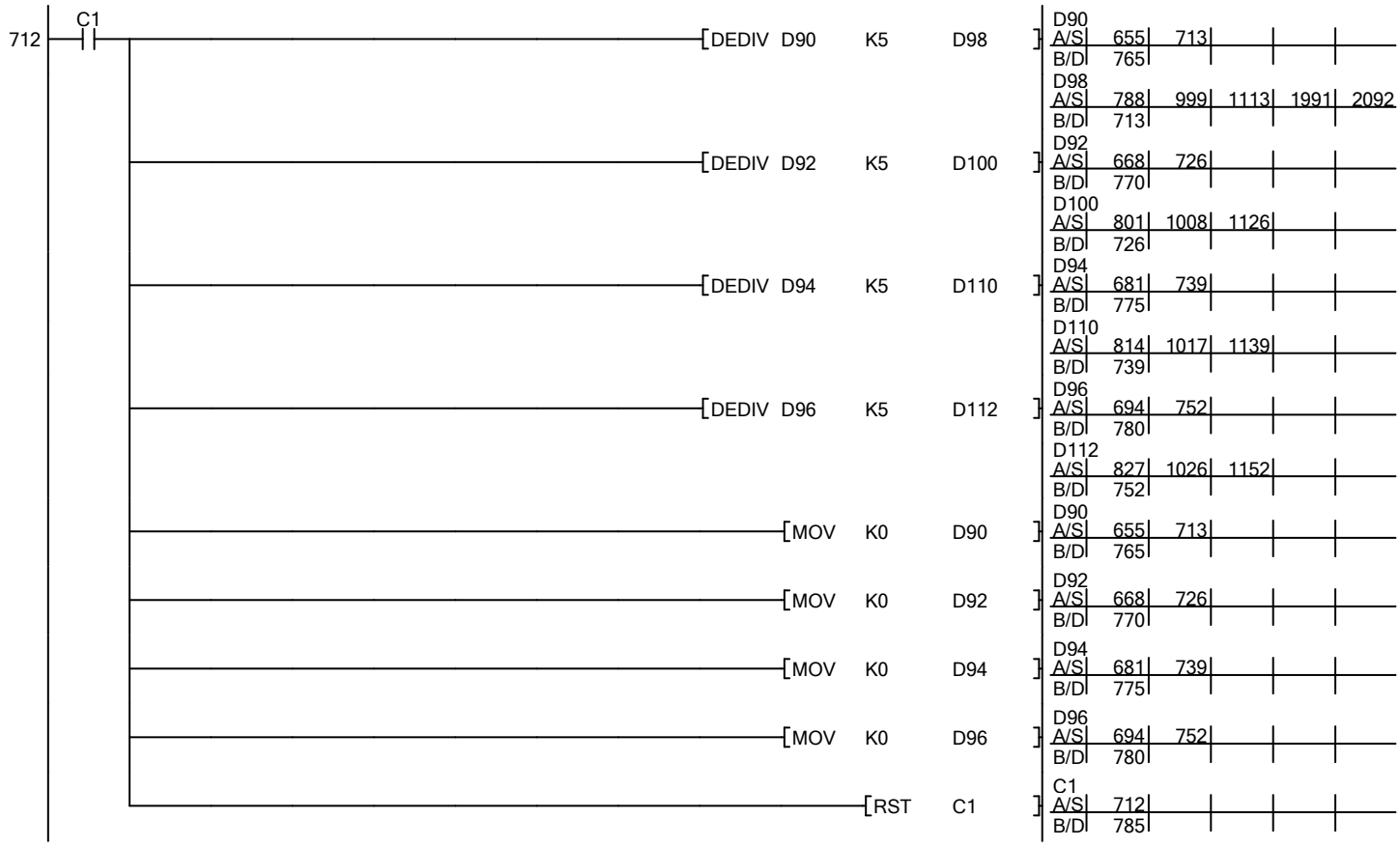

170
T0
DEMUL D0 E2.5 D22
D0 A/S 171 B/D 115
D22 A/S 249 B/D 171
DEMUL D2 E2.5 D24
D2 A/S 184 B/D 120
D24 A/S 262 B/D 184
DEMUL D4 E2.5 D26
D4 A/S 197 B/D 125
D26 A/S 275 B/D 197
DEMUL D6 E2.6 D28
D6 A/S 210 B/D 130
D28 A/S 288 B/D 210
DEMUL D8 E2.5 D30
D8 A/S 223 B/D 135
D30 A/S 301 B/D 223
DEMUL D10 E2.5 D32
D10 A/S 236 B/D 140
D32 A/S 314 B/D 236
DESUB D22 E1 D34
D22 A/S 249 B/D 171
D34 A/S 460 B/D 249
DESUB D24 E1 D36
D24 A/S 262 B/D 184
D36 A/S 473 B/D 262
DESUB D26 E1 D38
D26 A/S 275 B/D 197
D38 A/S 486 B/D 275
DESUB D28 E1 D40
D28 A/S 288 B/D 210
D40 A/S 499 B/D 288
DESUB D30 E1 D42
D30 A/S 301 B/D 223
D42 A/S 512 3460 B/D 301 3445
DESUB D32 E1 D44
D32 A/S 314 B/D 236
D44 A/S 525 B/D 314

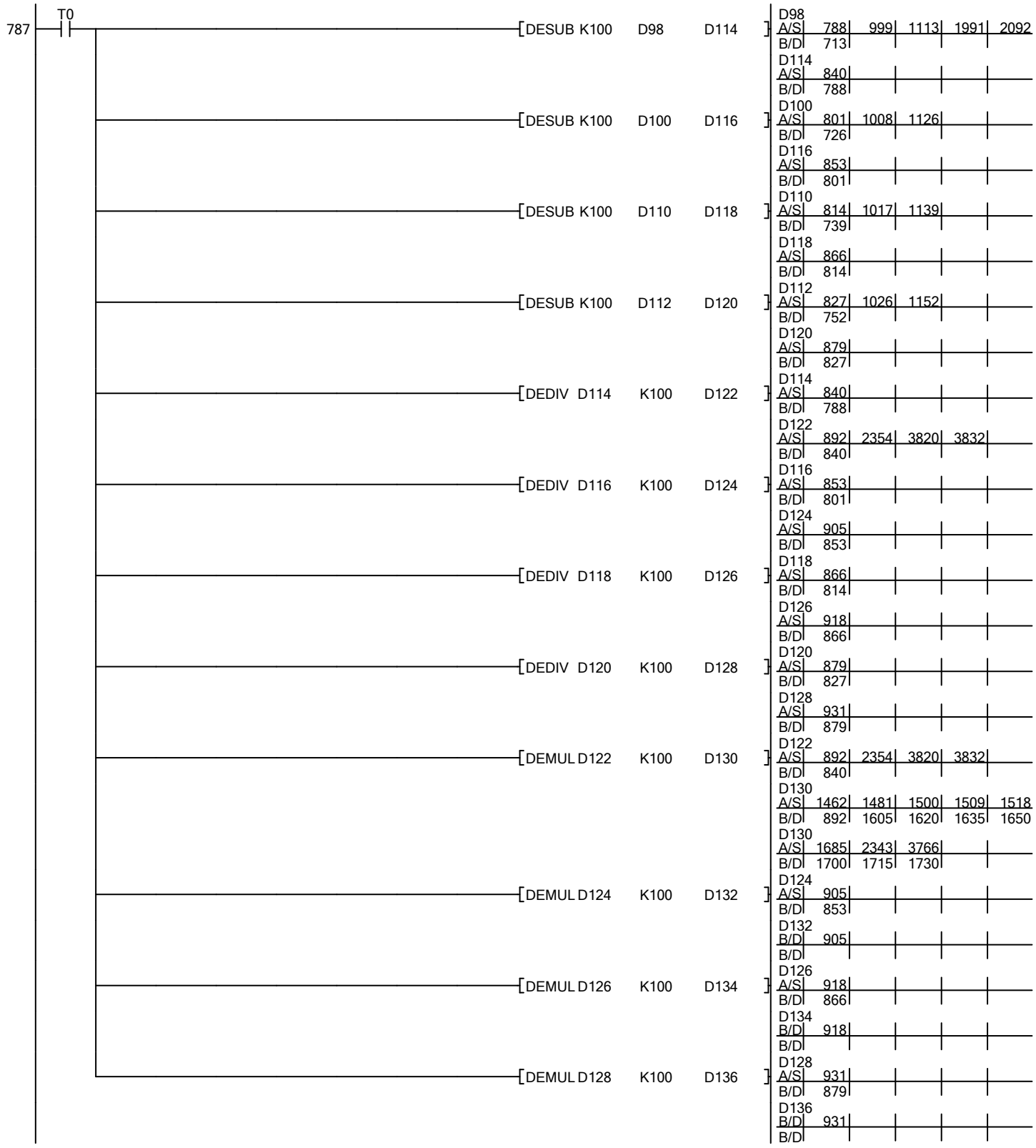

327
T0
DEMUL D12 E2.5 D46
D12 A/S 328 B/D 145
D46 A/S 380 B/D 328
DEMUL D14 E2.5 D48
D14 A/S 341 B/D 150
D48 A/S 393 B/D 341
DEMUL D16 E2.5 D50
D16 A/S 354 B/D 155
D50 A/S 406 B/D 354
DEMUL D18 E2.5 D52
D18 A/S 367 B/D 160
D52 A/S 419 B/D 367
DESUB D46 E1 D54
D46 A/S 380 B/D 328
D54 A/S 655 B/D 380
DESUB D48 E1 D56
D48 A/S 393 B/D 341
D56 A/S 668 B/D 393
DESUB D50 E1 D58
D50 A/S 406 B/D 354
D58 A/S 681 B/D 406
DESUB D52 E1 D60
D52 A/S 419 B/D 367
D60 A/S 694 B/D 419
432
T0
DEMUL D20 E2.5 D62
D20 A/S 433 B/D 165
D62 A/S 446 B/D 433
DESUB D62 E1 D64
D62 A/S 446 B/D 433
D64 A/S 1822 B/D 446

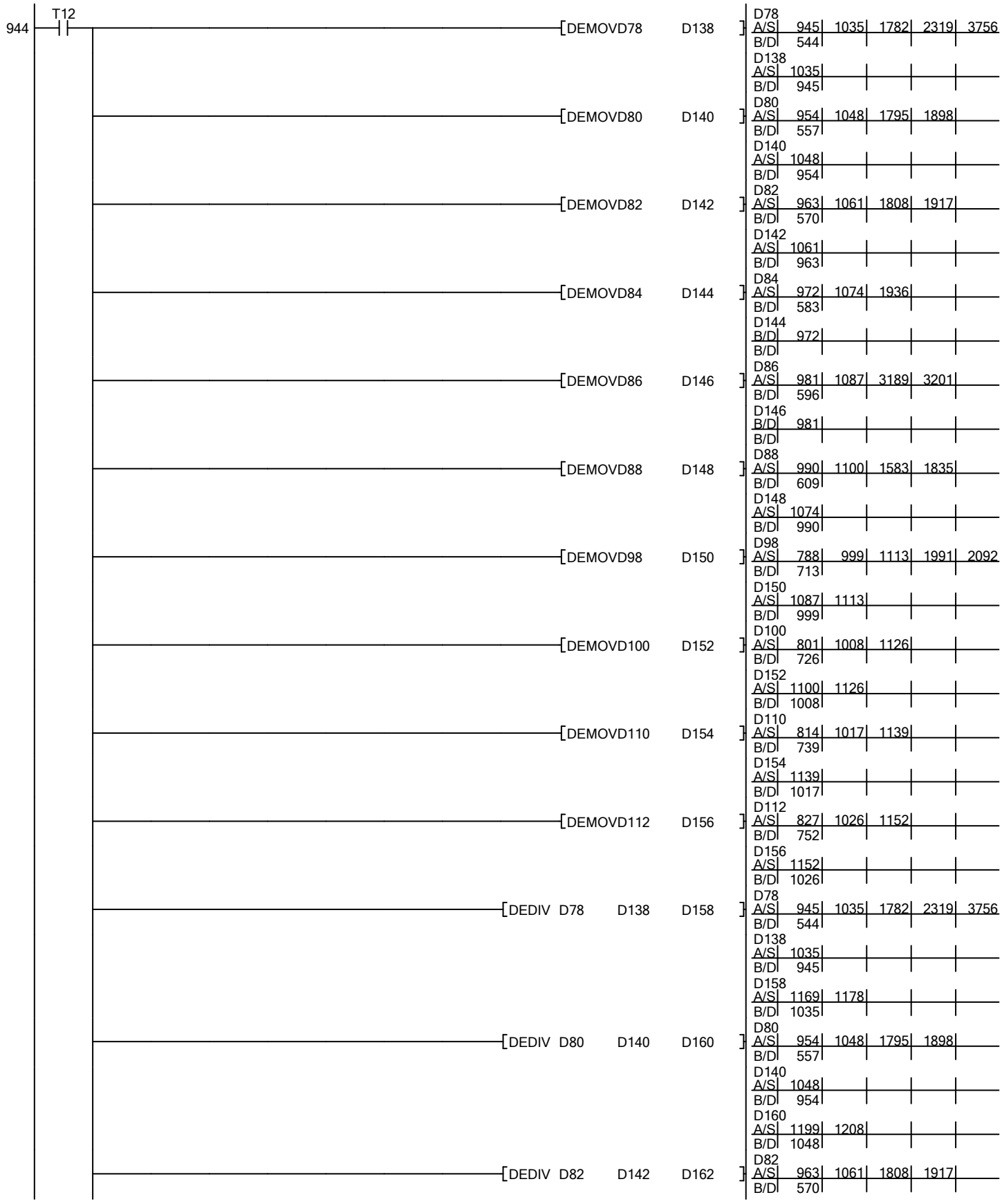

459
T10
DEADD D34 D66 D66
D34 A/S 460 B/D 249
D66 A/S 460 544 2176 3790 B/D 54 622
D66 A/S 460 544 2176 3790 B/D 54 622
DEADD D36 D68 D68
D36 A/S 473 B/D 262
D68 A/S 473 557 2218 3799 B/D 627
D68 A/S 473 557 2218 3799 B/D 627
DEADD D38 D70 D70
D38 A/S 486 B/D 275
D70 A/S 486 570 2260 3808 B/D 632
D70 A/S 486 570 2260 3808 B/D 632
DEADD D40 D72 D72
D40 A/S 499 B/D 288
D72 A/S 499 583 B/D 637
D72 A/S 499 583 B/D 637
DEADD D42 D74 D74
D42 A/S 512 3460 B/D 301 3445
D74 A/S 512 596 B/D 642
D74 A/S 512 596 B/D 642
DEADD D44 D76 D76
D44 A/S 525 B/D 314
D76 A/S 525 609 1769 B/D 647
D76 A/S 525 609 1769 B/D 647
K5
C0 C0
C0 A/S 543 B/D 652
RST T10
T10 A/S 459 B/D 541 3265

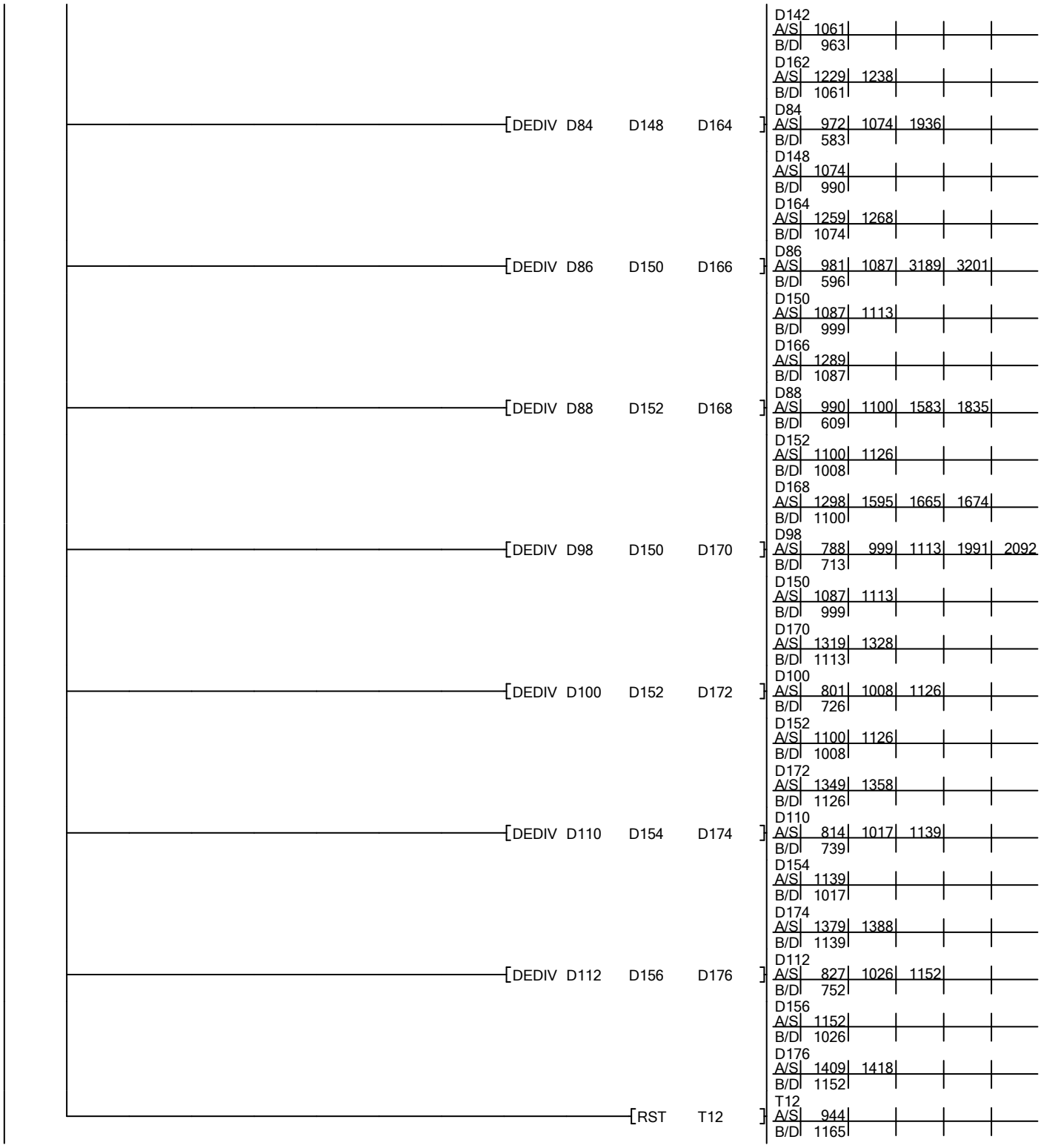

543
C0
DEDIV D66 K5 D78
DEDIV D68 K5 D80
DEDIV D70 K5 D82
DEDIV D72 K5 D84
DEDIV D74 K5 D86
DEDIV D76 K5 D88
MOV K0 D66
MOV K0 D68
MOV K0 D70
MOV K0 D72
MOV K0 D74
MOV K0 D76
RST C0
D66 A/S 460 544 2176 3790 B/D 54 622
D78 A/S 945 1035 1782 2319 3756 B/D 544
D68 A/S 473 557 2218 3799 B/D 627
D80 A/S 954 1048 1795 1898 B/D 557
D70 A/S 486 570 2260 3808 B/D 632
D82 A/S 963 1061 1808 1917 B/D 570
D72 A/S 499 583 B/D 637
D84 A/S 972 1074 1936 B/D 583
D74 A/S 512 596 B/D 642
D86 A/S 981 1087 3189 3201 B/D 596
D76 A/S 525 609 1769 B/D 647
D88 A/S 990 1100 1583 1835 B/D 609
D66 A/S 460 544 2176 3790 B/D 54 622
D68 A/S 473 557 2218 3799 B/D 627
D70 A/S 486 570 2260 3808 B/D 632
D72 A/S 499 583 B/D 637
D74 A/S 512 596 B/D 642
D76 A/S 525 609 1769 B/D 647
C0 A/S 543 B/D 652

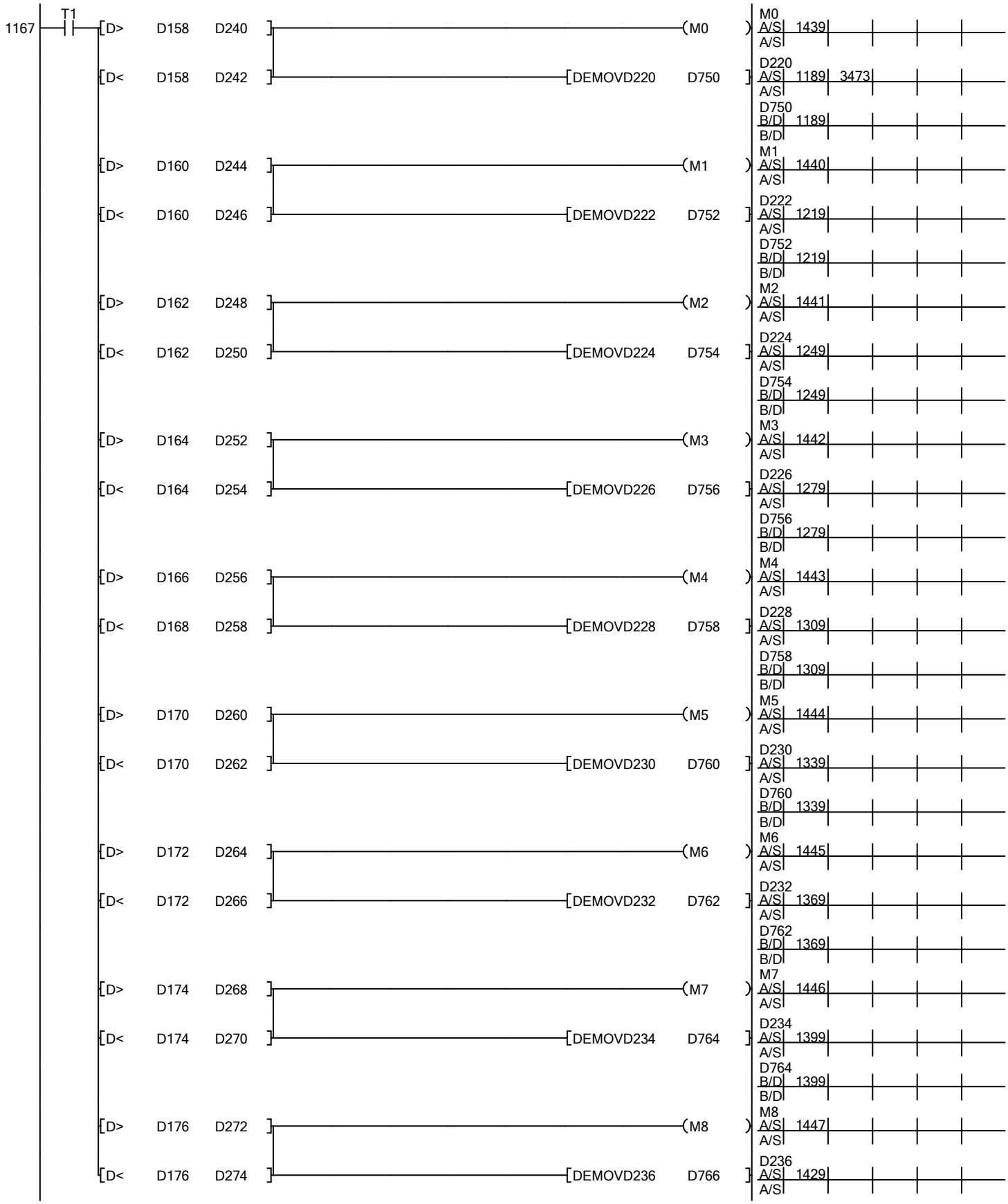

654
T11
DEADD D54 D90 D90
D54 A/S 655 B/D 380
D90 A/S 655 713 B/D 765
D90 A/S 655 713 B/D 765
DEADD D56 D92 D92
D56 A/S 668 B/D 393
D92 A/S 668 726 B/D 770
D92 A/S 668 726 B/D 770
DEADD D58 D94 D94
D58 A/S 681 B/D 406
D94 A/S 681 739 B/D 775
D94 A/S 681 739 B/D 775
DEADD D60 D96 D96
D60 A/S 694 B/D 419
D96 A/S 694 752 B/D 780
D96 A/S 694 752 B/D 780
K5
C1
C1 A/S 712 B/D 785
RST T11
T11 A/S 654 B/D 710 3267

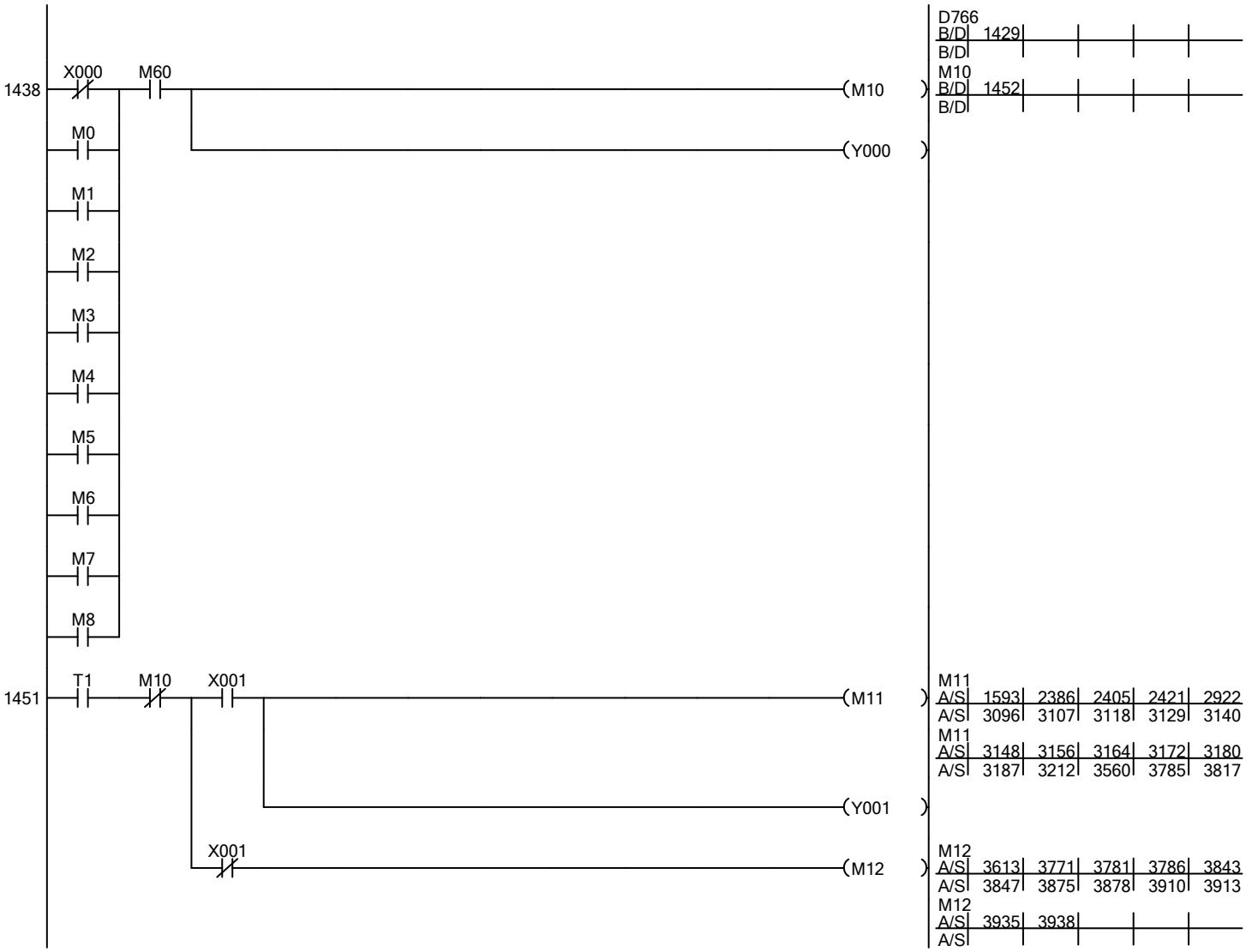

712
C1
DEDIV D90 K5 D98
DEDIV D92 K5 D100
DEDIV D94 K5 D110
DEDIV D96 K5 D112
MOV K0 D90
MOV K0 D92
MOV K0 D94
MOV K0 D96
RST C1
D90 A/S 655 713 B/D 765
D98 A/S 788 999 1113 1991 2092 B/D 713
D92 A/S 668 726 B/D 770
D100 A/S 801 1008 1126 B/D 726
D94 A/S 681 739 B/D 775
D110 A/S 814 1017 1139 B/D 739
D96 A/S 694 752 B/D 780
D112 A/S 827 1026 1152 B/D 752
D90 A/S 655 713 B/D 765
D92 A/S 668 726 B/D 770
D94 A/S 681 739 B/D 775
D96 A/S 694 752 B/D 780
C1 A/S 712 B/D 785

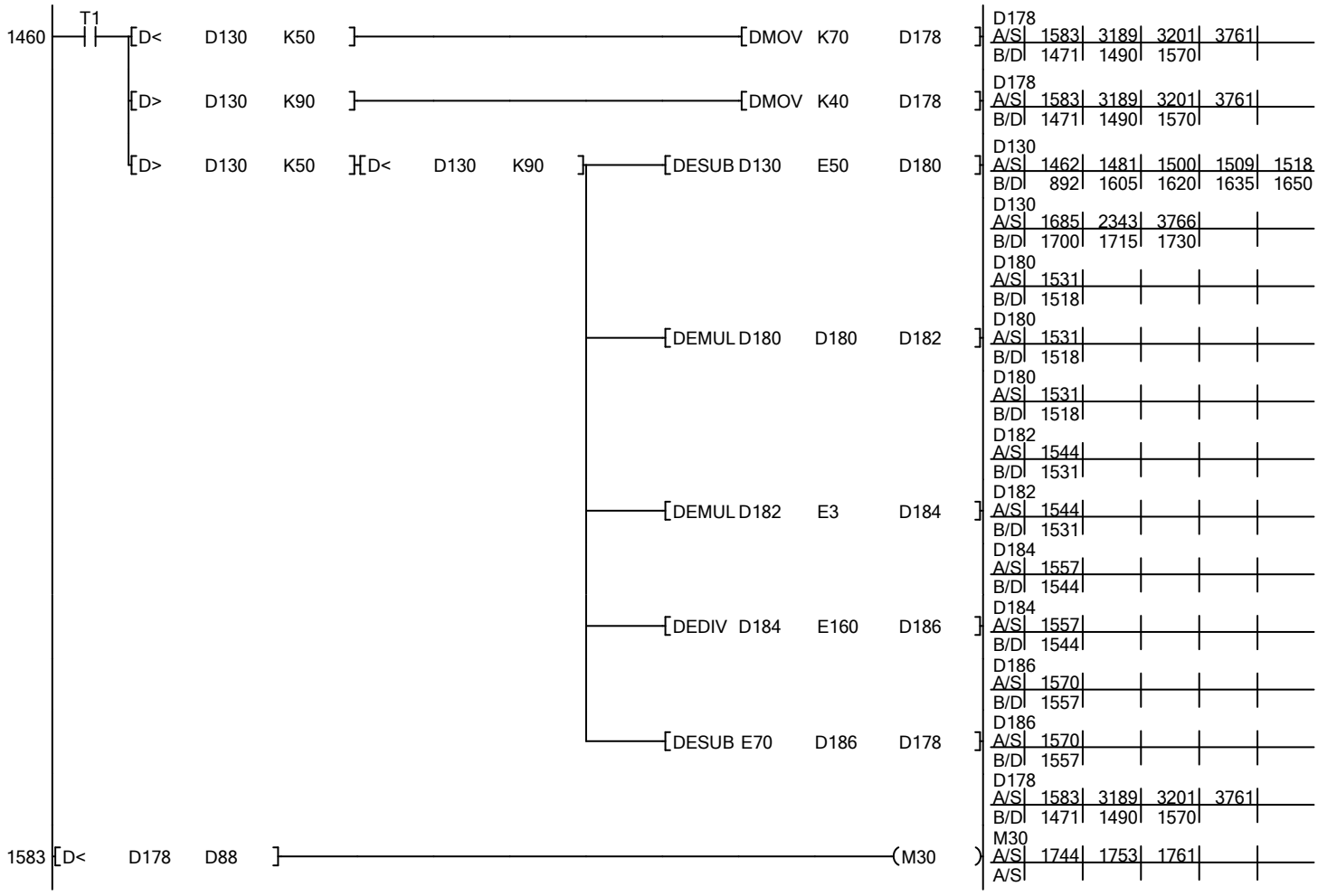

787
T0
DESUB K100 D98 D114
D98 A/S 788 999 1113 1991 2092 B/D 713
D114 A/S 840 B/D 788
DESUB K100 D100 D116
D100 A/S 801 1008 1126 B/D 726
D116 A/S 853 B/D 801
DESUB K100 D110 D118
D110 A/S 814 1017 1139 B/D 739
D118 A/S 866 B/D 814
DESUB K100 D112 D120
D112 A/S 827 1026 1152 B/D 752
D120 A/S 879 B/D 827
DEDIV D114 K100 D122
D114 A/S 840 B/D 788
D122 A/S 892 2354 3820 3832 B/D 840
DEDIV D116 K100 D124
D116 A/S 853 B/D 801
D124 A/S 905 B/D 853
DEDIV D118 K100 D126
D118 A/S 866 B/D 814
D126 A/S 918 B/D 866
DEDIV D120 K100 D128
D120 A/S 879 B/D 827
D128 A/S 931 B/D 879
DEMUL D122 K100 D130
D122 A/S 892 2354 3820 3832 B/D 840
D130 A/S 1462 1481 1500 1509 1518 B/D 892 1605 1620 1635 1650
D130 A/S 1685 2343 3766 B/D 1700 1715 1730
DEMUL D124 K100 D132
D124 A/S 905 B/D 853
D132 B/D 905 B/D
DEMUL D126 K100 D134
D126 A/S 918 B/D 866
D134 B/D 918 B/D
DEMUL D128 K100 D136
D128 A/S 931 B/D 879
D136 B/D 931 B/D

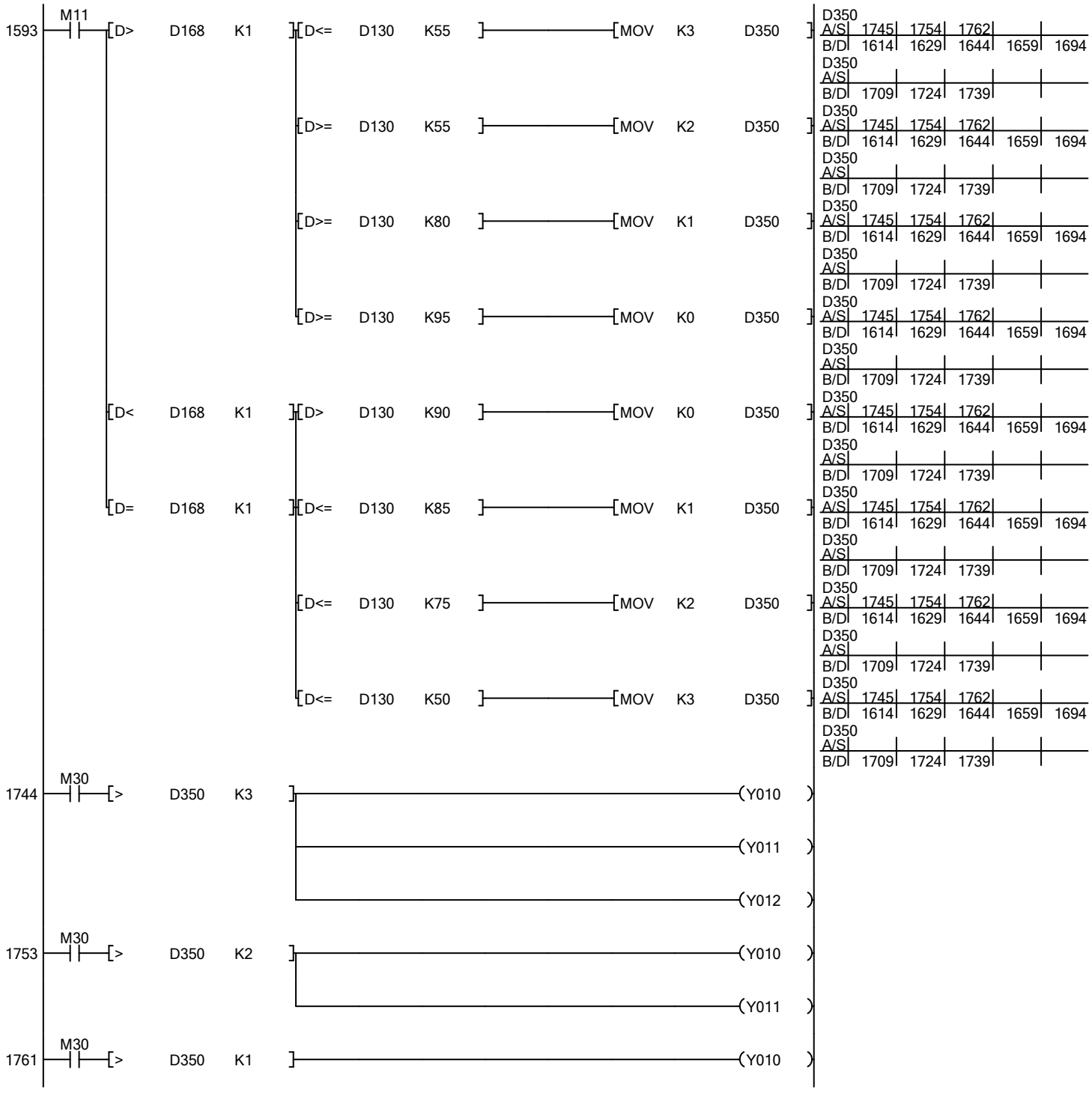

944
T12
DEMOVD78 D138
D78 A/S 945 1035 1782 2319 3756 B/D 544
D138 A/S 1035 B/D 945
DEMOVD80 D140
D80 A/S 954 1048 1795 1898 B/D 557
D140 A/S 1048 B/D 954
DEMOVD82 D142
D82 A/S 963 1061 1808 1917 B/D 570
D142 A/S 1061 B/D 963
DEMOVD84 D144
D84 A/S 972 1074 1936 B/D 583
D144 B/D 972 B/D
DEMOVD86 D146
D86 A/S 981 1087 3189 3201 B/D 596
D146 B/D 981 B/D
DEMOVD88 D148
D88 A/S 990 1100 1583 1835 B/D 609
D148 A/S 1074 B/D 990
DEMOVD98 D150
D98 A/S 788 999 1113 1991 2092 B/D 713
D150 A/S 1087 1113 B/D 999
DEMOVD100 D152
D100 A/S 801 1008 1126 B/D 726
D152 A/S 1100 1126 B/D 1008
DEMOVD110 D154
D110 A/S 814 1017 1139 B/D 739
D154 A/S 1139 B/D 1017
DEMOVD112 D156
D112 A/S 827 1026 1152 B/D 752
D156 A/S 1152 B/D 1026
DEDIV D78 D138 D158
D78 A/S 945 1035 1782 2319 3756 B/D 544
D138 A/S 1035 B/D 945
D158 A/S 1169 1178 B/D 1035
DEDIV D80 D140 D160
D80 A/S 954 1048 1795 1898 B/D 557
D140 A/S 1048 B/D 954
D160 A/S 1199 1208 B/D 1048
DEDIV D82 D142 D162
D82 A/S 963 1061 1808 1917 B/D 570

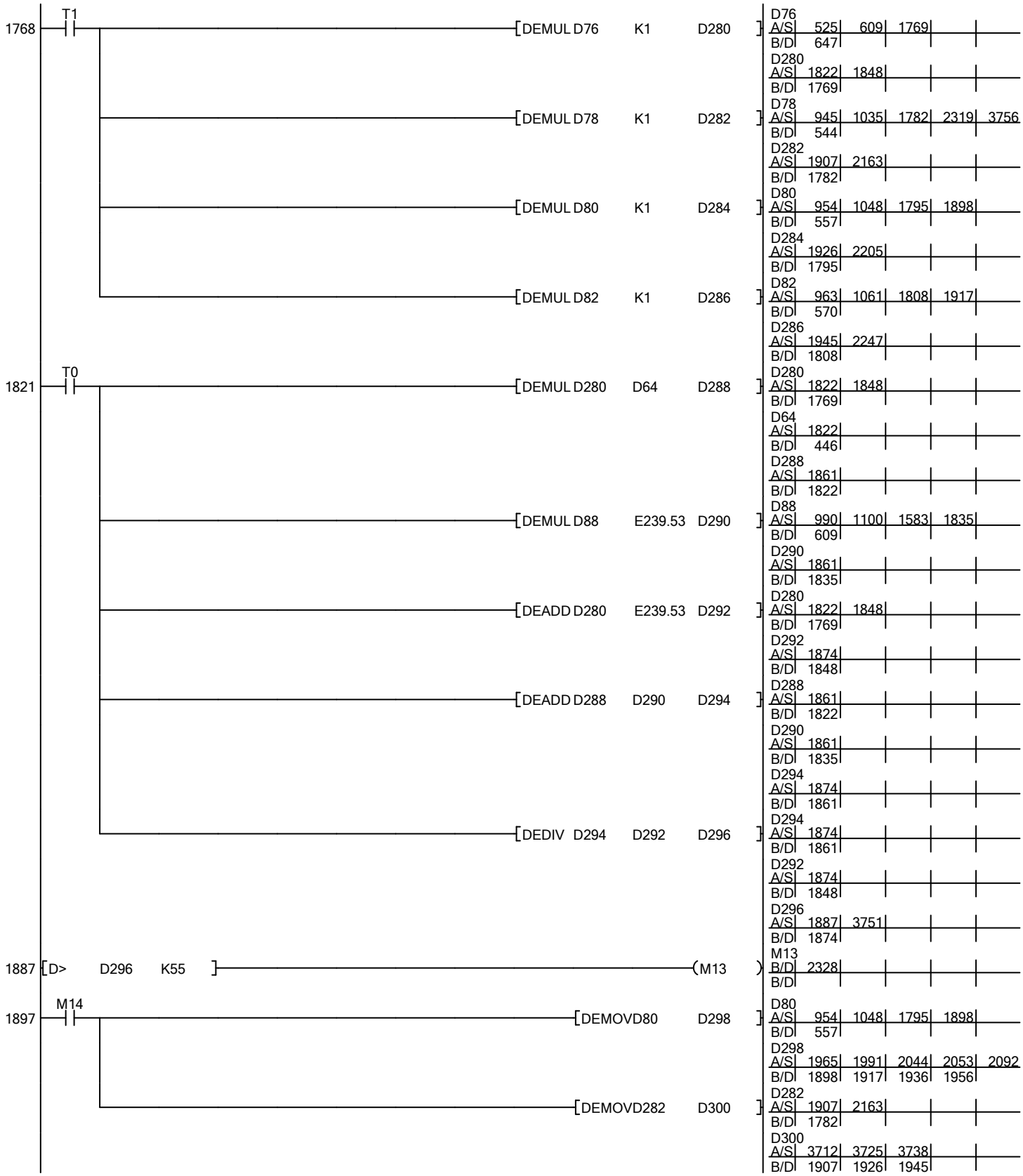

[DEDIV D84 D148 D164]
D142 A/S 1061 B/D 963
D162 A/S 1229 1238 B/D 1061
D84 A/S 972 1074 1936 B/D 583
D148 A/S 1074 B/D 990
D164 A/S 1259 1268 B/D 1074
[DEDIV D86 D150 D166]
D86 A/S 981 1087 3189 3201 B/D 596
D150 A/S 1087 1113 B/D 999
D166 A/S 1289 B/D 1087
[DEDIV D88 D152 D168]
D88 A/S 990 1100 1583 1835 B/D 609
D152 A/S 1100 1126 B/D 1008
D168 A/S 1298 1595 1665 1674 B/D 1100
[DEDIV D98 D150 D170]
D98 A/S 788 999 1113 1991 2092 B/D 713
D150 A/S 1087 1113 B/D 999
D170 A/S 1319 1328 B/D 1113
[DEDIV D100 D152 D172]
D100 A/S 801 1008 1126 B/D 726
D152 A/S 1100 1126 B/D 1008
D172 A/S 1349 1358 B/D 1126
[DEDIV D110 D154 D174]
D110 A/S 814 1017 1139 B/D 739
D154 A/S 1139 B/D 1017
D174 A/S 1379 1388 B/D 1139
[DEDIV D112 D156 D176]
D112 A/S 827 1026 1152 B/D 752
D156 A/S 1152 B/D 1026
D176 A/S 1409 1418 B/D 1152
[RST T12]
T12 A/S 944 B/D 1165

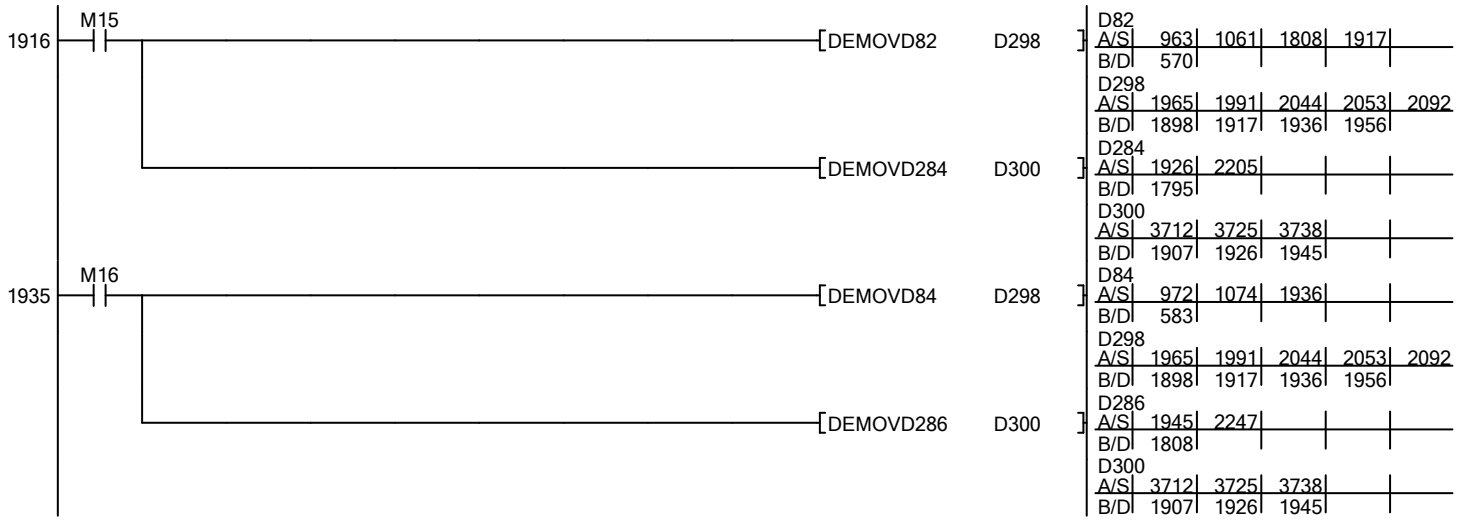

1167
T1
D> D158 D240 M0
D< D158 D242 DEMOVD220 D750
D> D160 D244 M1
D< D160 D246 DEMOVD222 D752
D> D162 D248 M2
D< D162 D250 DEMOVD224 D754
D> D164 D252 M3
D< D164 D254 DEMOVD226 D756
D> D166 D256 M4
D< D168 D258 DEMOVD228 D758
D> D170 D260 M5
D< D170 D262 DEMOVD230 D760
D> D172 D264 M6
D< D172 D266 DEMOVD232 D762
D> D174 D268 M7
D< D174 D270 DEMOVD234 D764
D> D176 D272 M8
D< D176 D274 DEMOVD236 D766
M0 A/S 1439 A/S
D220 A/S 1189 3473 A/S
D750 B/D 1189 B/D
M1 A/S 1440 A/S
D222 A/S 1219 A/S
D752 B/D 1219 B/D
M2 A/S 1441 A/S
D224 A/S 1249 A/S
D754 B/D 1249 B/D
M3 A/S 1442 A/S
D226 A/S 1279 A/S
D756 B/D 1279 B/D
M4 A/S 1443 A/S
D228 A/S 1309 A/S
D758 B/D 1309 B/D
M5 A/S 1444 A/S
D230 A/S 1339 A/S
D760 B/D 1339 B/D
M6 A/S 1445 A/S
D232 A/S 1369 A/S
D762 B/D 1369 B/D
M7 A/S 1446 A/S
D234 A/S 1399 A/S
D764 B/D 1399 B/D
M8 A/S 1447 A/S
D236 A/S 1429 A/S

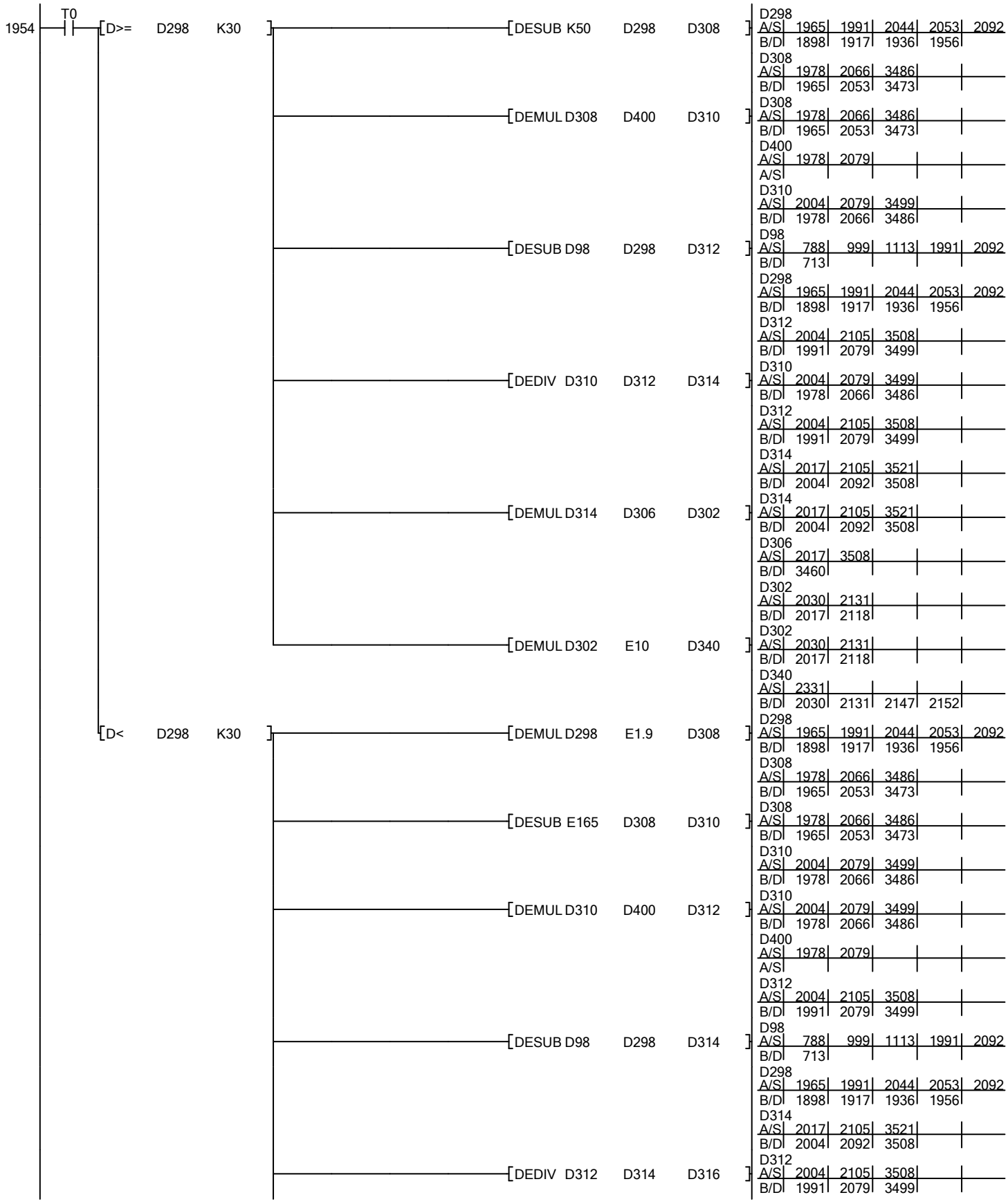

D766
B/D 1429
B/D
M10
B/D 1452
B/D
1438
X000
M60
M10
Y000
M0
M1
M2
M3
M4
M5
M6
M7
M8
1451
T1
M10
X001
M11
Y001
X001
M12
M11
A/S 1593 2386 2405 2421 2922
A/S 3096 3107 3118 3129 3140
M11
A/S 3148 3156 3164 3172 3180
A/S 3187 3212 3560 3785 3817
M12
A/S 3613 3771 3781 3786 3843
A/S 3847 3875 3878 3910 3913
M12
A/S 3935 3938
A/S

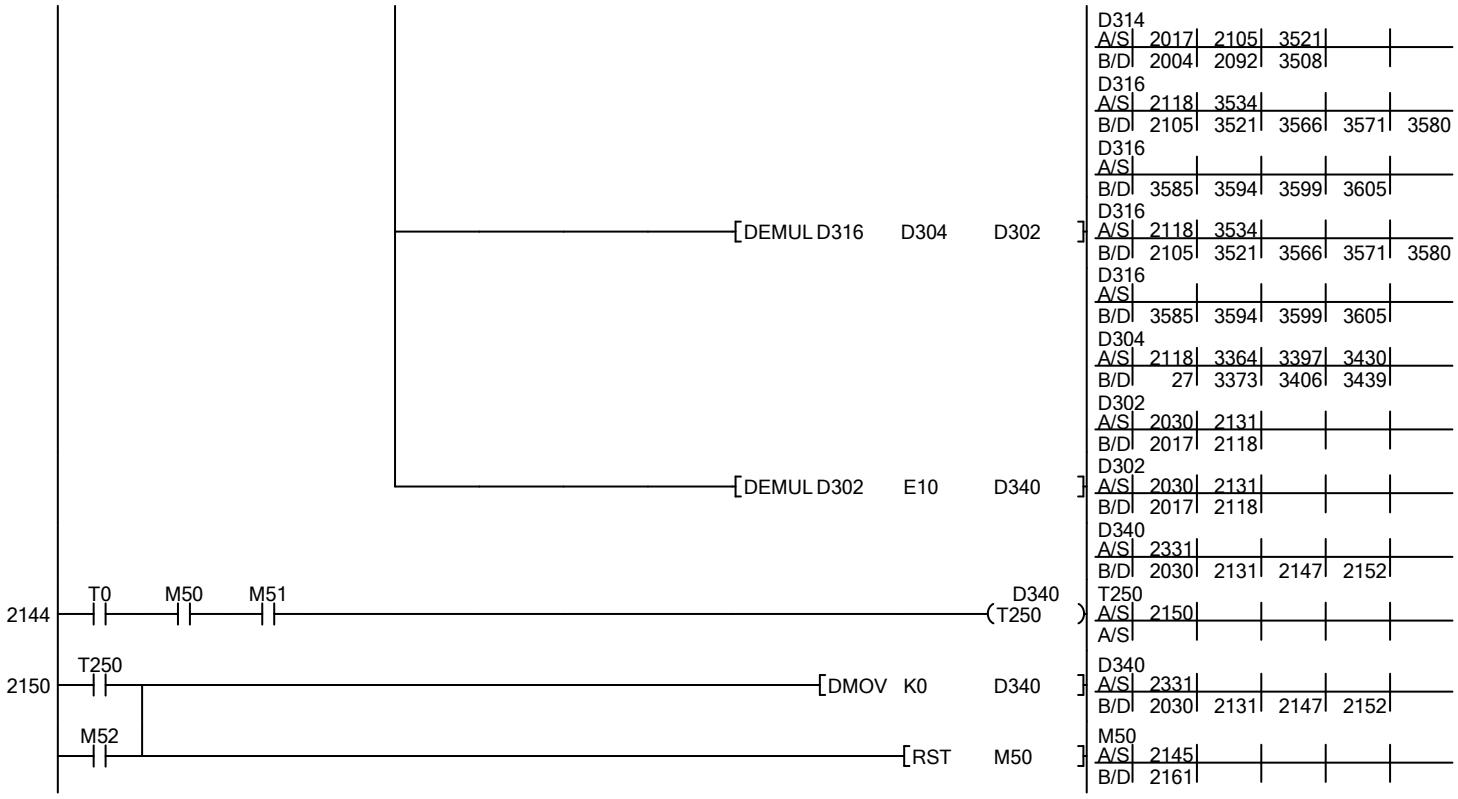

1460
T1
D< D130 K50
D> D130 K90
D> D130 K50
D< D130 K90
DMOV K70 D178
DMOV K40 D178
DESUB D130 E50 D180
DEMUL D180 D180 D182
DEMUL D182 E3 D184
DEDIV D184 E160 D186
DESUB E70 D186 D178
1583
D< D178 D88
M30
D178 A/S 1583 3189 3201 3761 B/D 1471 1490 1570
D130 A/S 1462 1481 1500 1509 1518 B/D 892 1605 1620 1635 1650
D130 A/S 1685 2343 3766 B/D 1700 1715 1730
D180 A/S 1531 B/D 1518
D182 A/S 1544 B/D 1531
D184 A/S 1557 B/D 1544
D186 A/S 1570 B/D 1557
M30 A/S 1744 1753 1761 A/S

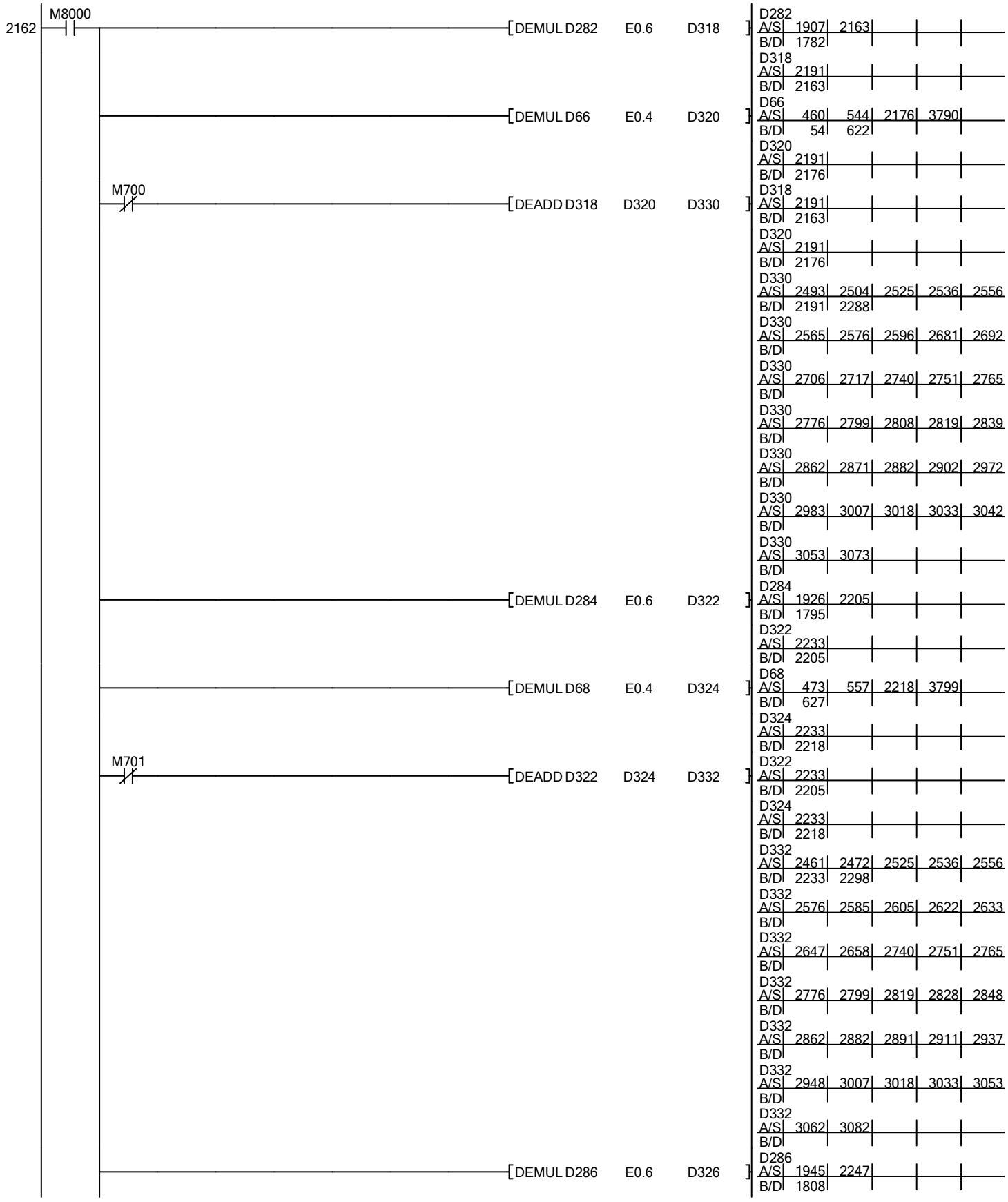

1593 M11 [D> D168 K1]
[D<= D130 K55] [MOV K3 D350]
[D>= D130 K55] [MOV K2 D350]
[D>= D130 K80] [MOV K1 D350]
[D>= D130 K95] [MOV K0 D350]
[D< D168 K1]
[D> D130 K90] [MOV K0 D350]
[D= D168 K1]
[D<= D130 K85] [MOV K1 D350]
[D<= D130 K75] [MOV K2 D350]
[D<= D130 K50] [MOV K3 D350]
D350 A/S 1745 1754 1762
B/D 1614 1629 1644 1659 1694
D350 A/S
B/D 1709 1724 1739
1744 M30 [> D350 K3]
(Y010)
(Y011)
(Y012)
1753 M30 [> D350 K2]
(Y010)
(Y011)
1761 M30 [> D350 K1]
(Y010)

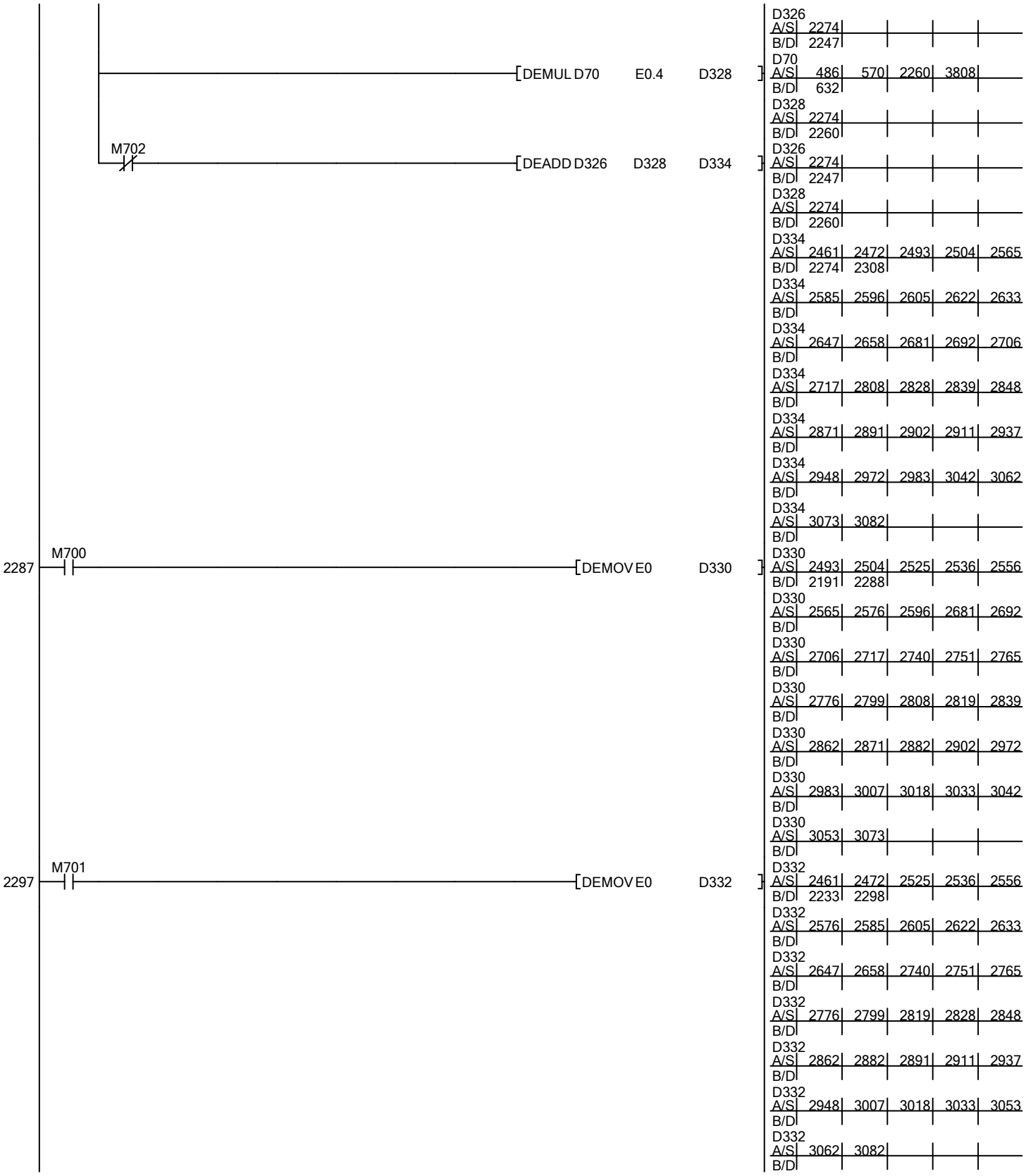

1768 T1
[DEMUL D76 K1 D280]
[DEMUL D78 K1 D282]
[DEMUL D80 K1 D284]
[DEMUL D82 K1 D286]
1821 T0
[DEMUL D280 D64 D288]
[DEMUL D88 E239.53 D290]
[DEADD D280 E239.53 D292]
[DEADD D288 D290 D294]
[DEDIV D294 D292 D296]
1887 [D> D296 K55]
(M13)
1897 M14
[DEMOVD80 D298]
[DEMOVD282 D300]
D76 A/S 525 609 1769 B/D 647
D280 A/S 1822 1848 B/D 1769
D78 A/S 945 1035 1782 2319 3756 B/D 544
D282 A/S 1907 2163 B/D 1782
D80 A/S 954 1048 1795 1898 B/D 557
D284 A/S 1926 2205 B/D 1795
D82 A/S 963 1061 1808 1917 B/D 570
D286 A/S 1945 2247 B/D 1808
D280 A/S 1822 1848 B/D 1769
D64 A/S 1822 B/D 446
D288 A/S 1861 B/D 1822
D88 A/S 990 1100 1583 1835 B/D 609
D290 A/S 1861 B/D 1835
D280 A/S 1822 1848 B/D 1769
D292 A/S 1874 B/D 1848
D288 A/S 1861 B/D 1822
D290 A/S 1861 B/D 1835
D294 A/S 1874 B/D 1861
D294 A/S 1874 B/D 1861
D292 A/S 1874 B/D 1848
D296 A/S 1887 3751 B/D 1874
M13 B/D 2328 B/D
D80 A/S 954 1048 1795 1898 B/D 557
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D282 A/S 1907 2163 B/D 1782
D300 A/S 3712 3725 3738 B/D 1907 1926 1945

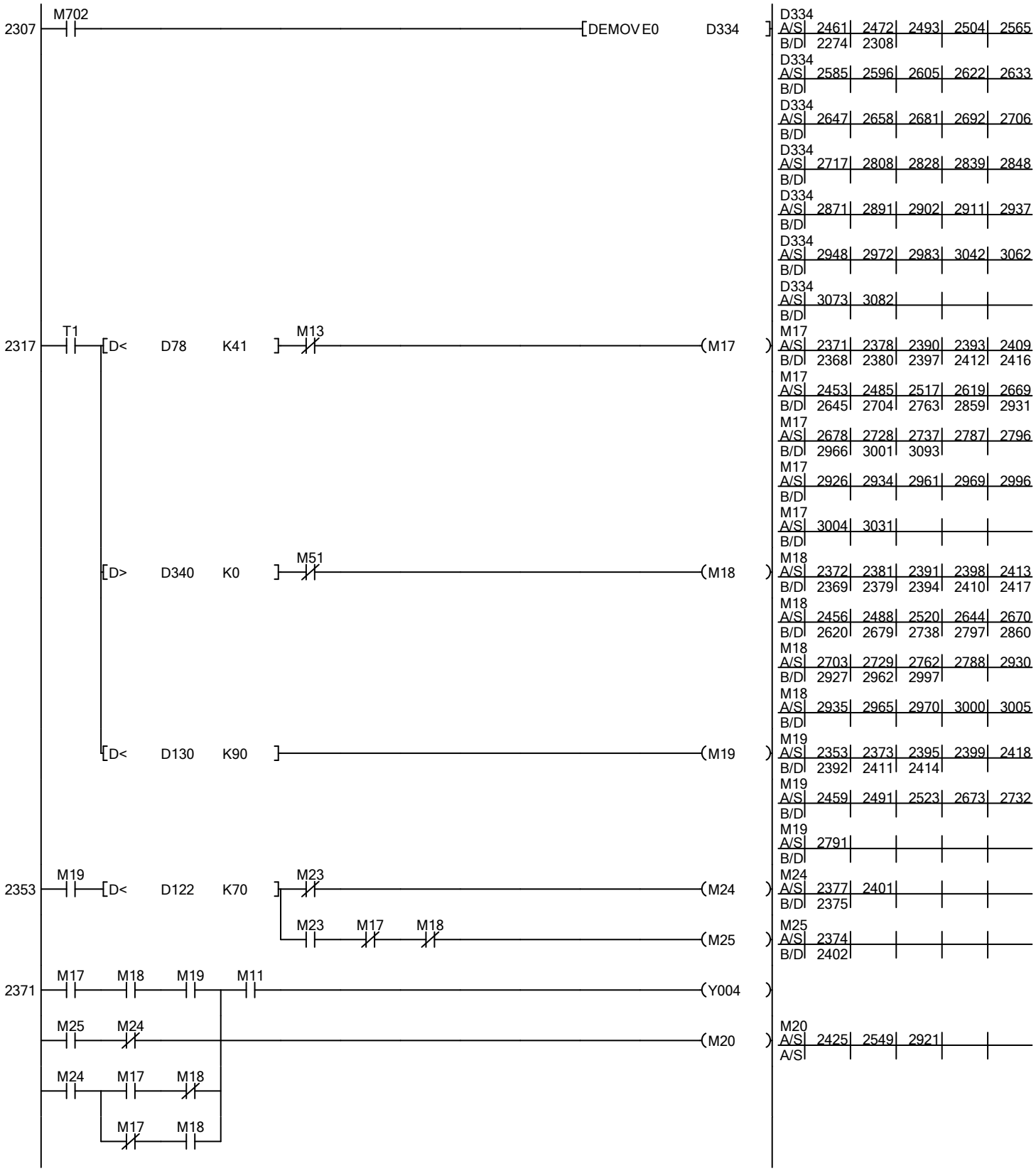

M15
1916
DEMOVD82
D298
DEMOVD284
D300
M16
1935
DEMOVD84
D298
DEMOVD286
D300
D82
A/S 963 1061 1808 1917
B/D 570
D298
A/S 1965 1991 2044 2053 2092
B/D 1898 1917 1936 1956
D284
A/S 1926 2205
B/D 1795
D300
A/S 3712 3725 3738
B/D 1907 1926 1945
D84
A/S 972 1074 1936
B/D 583
D298
A/S 1965 1991 2044 2053 2092
B/D 1898 1917 1936 1956
D286
A/S 1945 2247
B/D 1808
D300
A/S 3712 3725 3738
B/D 1907 1926 1945

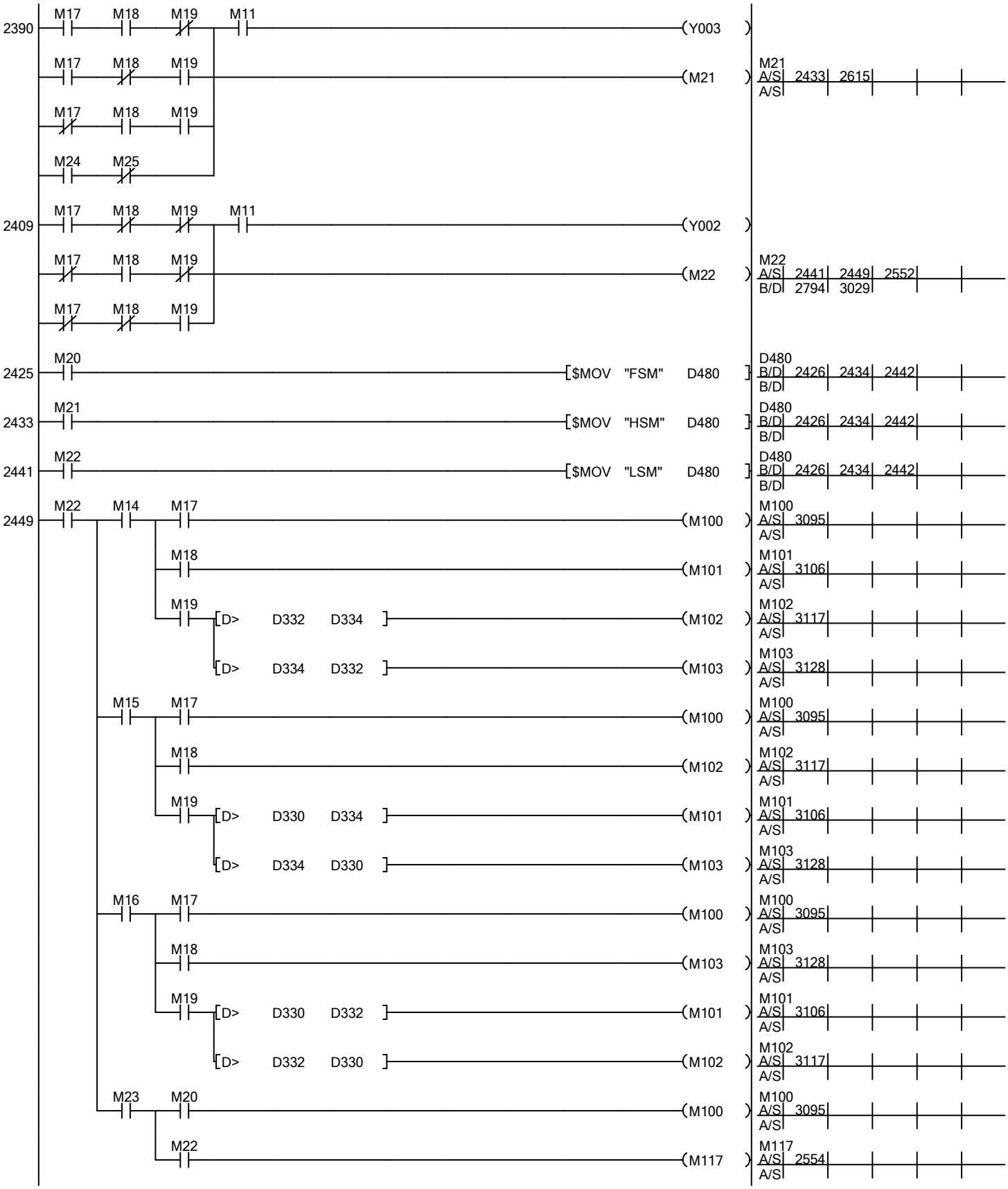

1954
T0
D>= D298 K30
DESUB K50 D298 D308
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D308 A/S 1978 2066 3486 B/D 1965 2053 3473
DEMUL D308 D400 D310
D308 A/S 1978 2066 3486 B/D 1965 2053 3473
D400 A/S 1978 2079 A/S
D310 A/S 2004 2079 3499 B/D 1978 2066 3486
DESUB D98 D298 D312
D98 A/S 788 999 1113 1991 2092 B/D 713
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D312 A/S 2004 2105 3508 B/D 1991 2079 3499
DEDIV D310 D312 D314
D310 A/S 2004 2079 3499 B/D 1978 2066 3486
D312 A/S 2004 2105 3508 B/D 1991 2079 3499
D314 A/S 2017 2105 3521 B/D 2004 2092 3508
DEMUL D314 D306 D302
D314 A/S 2017 2105 3521 B/D 2004 2092 3508
D306 A/S 2017 3508 B/D 3460
D302 A/S 2030 2131 B/D 2017 2118
DEMUL D302 E10 D340
D302 A/S 2030 2131 B/D 2017 2118
D340 A/S 2331 B/D 2030 2131 2147 2152
D< D298 K30
DEMUL D298 E1.9 D308
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D308 A/S 1978 2066 3486 B/D 1965 2053 3473
DESUB E165 D308 D310
D308 A/S 1978 2066 3486 B/D 1965 2053 3473
D310 A/S 2004 2079 3499 B/D 1978 2066 3486
DEMUL D310 D400 D312
D310 A/S 2004 2079 3499 B/D 1978 2066 3486
D400 A/S 1978 2079 A/S
D312 A/S 2004 2105 3508 B/D 1991 2079 3499
DESUB D98 D298 D314
D98 A/S 788 999 1113 1991 2092 B/D 713
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D314 A/S 2017 2105 3521 B/D 2004 2092 3508
DEDIV D312 D314 D316
D312 A/S 2004 2105 3508 B/D 1991 2079 3499

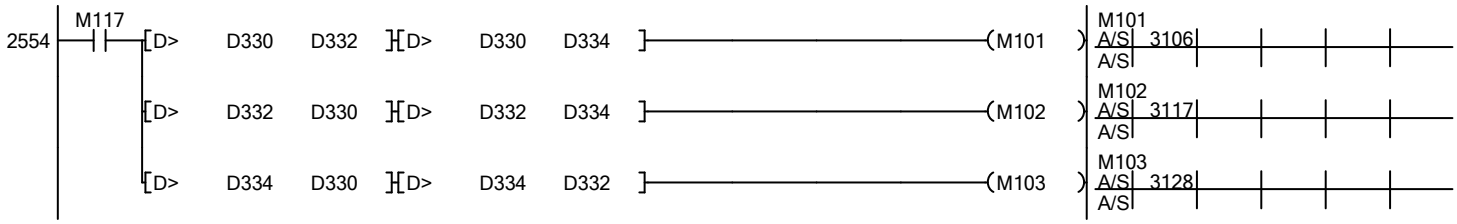

DEMUL D316 D304 D302
DEMUL D302 E10 D340
2144
T0 M50 M51
D340
T250
2150
T250
DMOV K0 D340
M52
RST M50
D314 A/S 2017 2105 3521 B/D 2004 2092 3508
D316 A/S 2118 3534 B/D 2105 3521 3566 3571 3580
D316 A/S B/D 3585 3594 3599 3605
D316 A/S 2118 3534 B/D 2105 3521 3566 3571 3580
D316 A/S B/D 3585 3594 3599 3605
D304 A/S 2118 3364 3397 3430 B/D 27 3373 3406 3439
D302 A/S 2030 2131 B/D 2017 2118
D302 A/S 2030 2131 B/D 2017 2118
D340 A/S 2331 B/D 2030 2131 2147 2152
T250 A/S 2150 A/S
D340 A/S 2331 B/D 2030 2131 2147 2152
M50 A/S 2145 B/D 2161

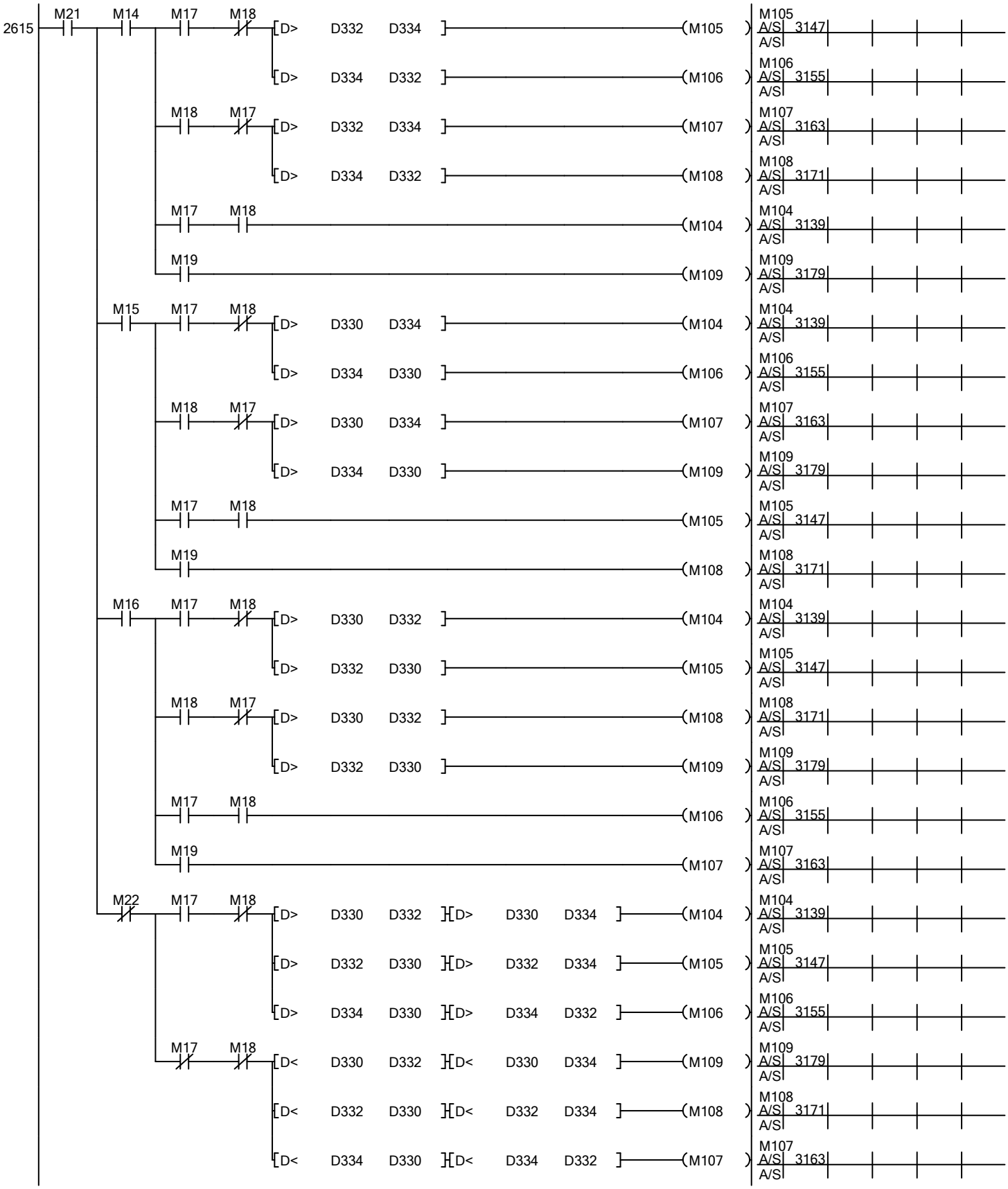

2162
M8000
DEMUL D282 E0.6 D318
D282 A/S 1907 2163 B/D 1782
D318 A/S 2191 B/D 2163
DEMUL D66 E0.4 D320
D66 A/S 460 544 2176 3790 B/D 54 622
D320 A/S 2191 B/D 2176
M700
DEADD D318 D320 D330
D318 A/S 2191 B/D 2163
D320 A/S 2191 B/D 2176
D330 A/S 2493 2504 2525 2536 2556 B/D 2191 2288
D330 A/S 2565 2576 2596 2681 2692 B/D
D330 A/S 2706 2717 2740 2751 2765 B/D
D330 A/S 2776 2799 2808 2819 2839 B/D
D330 A/S 2862 2871 2882 2902 2972 B/D
D330 A/S 2983 3007 3018 3033 3042 B/D
D330 A/S 3053 3073 B/D
DEMUL D284 E0.6 D322
D284 A/S 1926 2205 B/D 1795
D322 A/S 2233 B/D 2205
DEMUL D68 E0.4 D324
D68 A/S 473 557 2218 3799 B/D 627
D324 A/S 2233 B/D 2218
M701
DEADD D322 D324 D332
D322 A/S 2233 B/D 2205
D324 A/S 2233 B/D 2218
D332 A/S 2461 2472 2525 2536 2556 B/D 2233 2298
D332 A/S 2576 2585 2605 2622 2633 B/D
D332 A/S 2647 2658 2740 2751 2765 B/D
D332 A/S 2776 2799 2819 2828 2848 B/D
D332 A/S 2862 2882 2891 2911 2937 B/D
D332 A/S 2948 3007 3018 3033 3053 B/D
D332 A/S 3062 3082 B/D
DEMUL D286 E0.6 D326
D286 A/S 1945 2247 B/D 1808

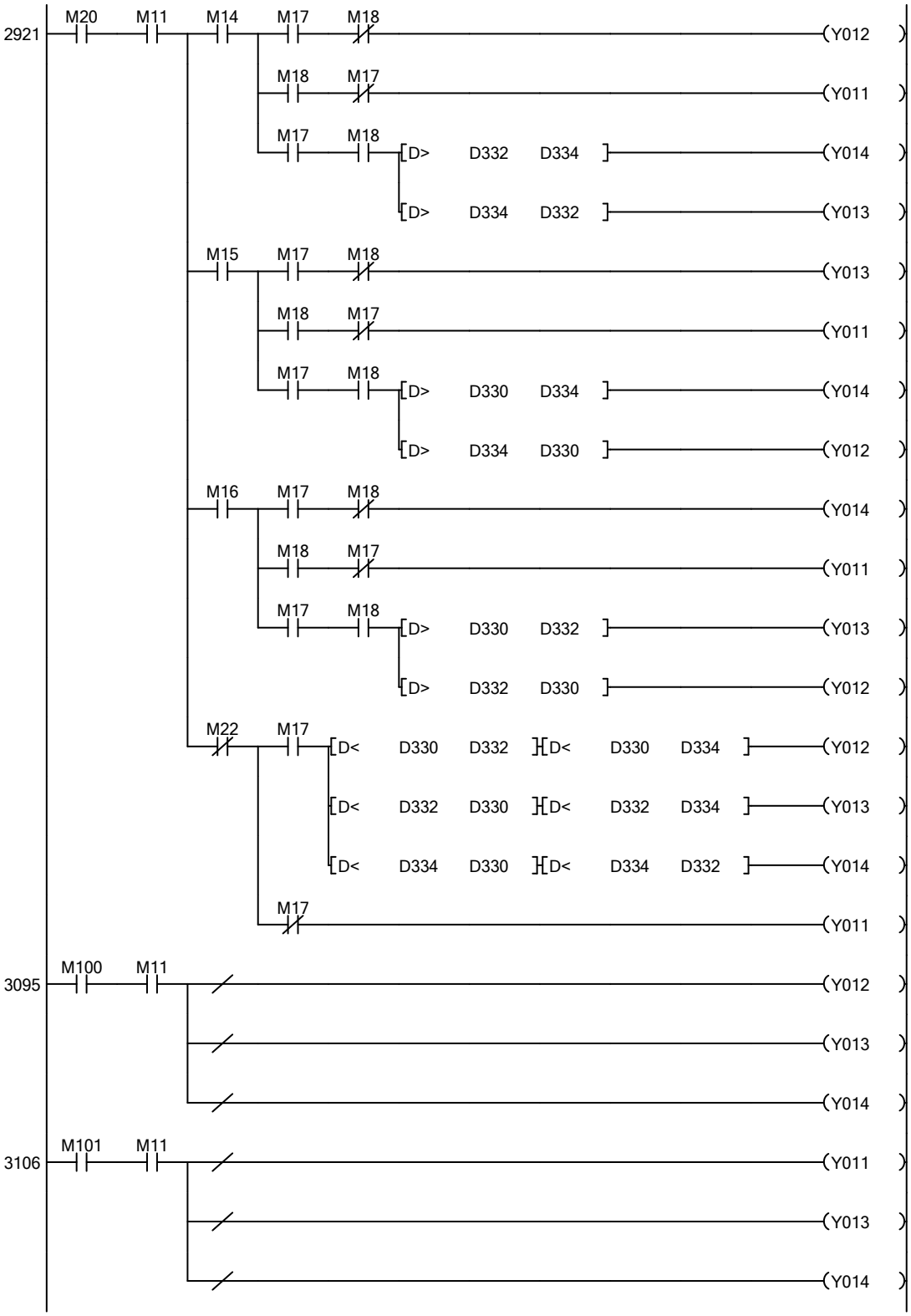

DEMUL D70 E0.4 D328
M702
DEADD D326 D328 D334
2287
M700
DEMOV E0 D330
2297
M701
DEMOV E0 D332
D326 A/S 2274 B/D 2247
D70 A/S 486 570 2260 3808 B/D 632
D328 A/S 2274 B/D 2260
D326 A/S 2274 B/D 2247
D328 A/S 2274 B/D 2260
D334 A/S 2461 2472 2493 2504 2565 B/D 2274 2308
D334 A/S 2585 2596 2605 2622 2633 B/D
D334 A/S 2647 2658 2681 2692 2706 B/D
D334 A/S 2717 2808 2828 2839 2848 B/D
D334 A/S 2871 2891 2902 2911 2937 B/D
D334 A/S 2948 2972 2983 3042 3062 B/D
D334 A/S 3073 3082 B/D
D330 A/S 2493 2504 2525 2536 2556 B/D 2191 2288
D330 A/S 2565 2576 2596 2681 2692 B/D
D330 A/S 2706 2717 2740 2751 2765 B/D
D330 A/S 2776 2799 2808 2819 2839 B/D
D330 A/S 2862 2871 2882 2902 2972 B/D
D330 A/S 2983 3007 3018 3033 3042 B/D
D330 A/S 3053 3073 B/D
D332 A/S 2461 2472 2525 2536 2556 B/D 2233 2298
D332 A/S 2576 2585 2605 2622 2633 B/D
D332 A/S 2647 2658 2740 2751 2765 B/D
D332 A/S 2776 2799 2819 2828 2848 B/D
D332 A/S 2862 2882 2891 2911 2937 B/D
D332 A/S 2948 3007 3018 3033 3053 B/D
D332 A/S 3062 3082 B/D

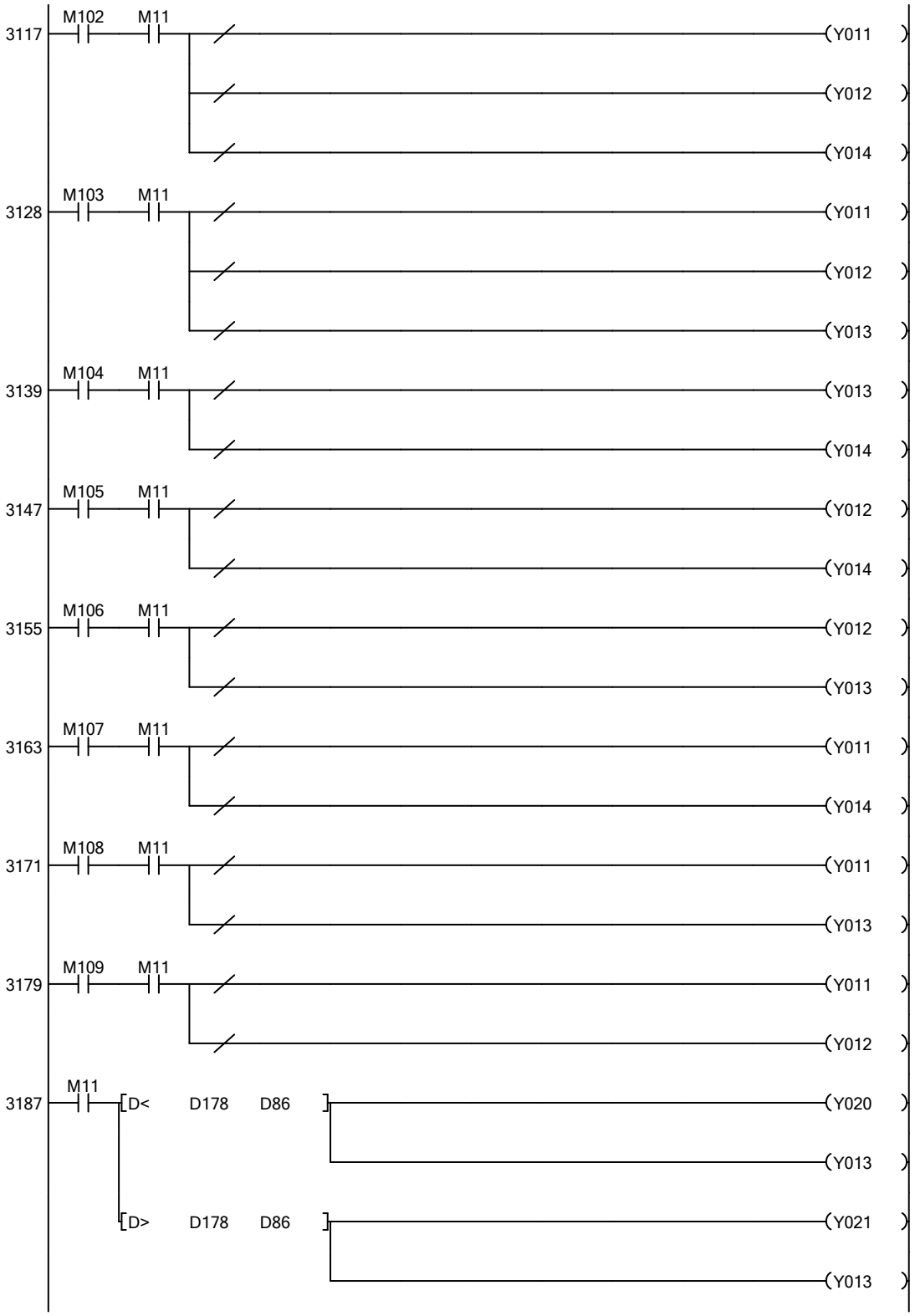

2307
M702
DEMOVE0 D334
D334 A/S 2461 2472 2493 2504 2565 B/D 2274 2308
D334 A/S 2585 2596 2605 2622 2633 B/D
D334 A/S 2647 2658 2681 2692 2706 B/D
D334 A/S 2717 2808 2828 2839 2848 B/D
D334 A/S 2871 2891 2902 2911 2937 B/D
D334 A/S 2948 2972 2983 3042 3062 B/D
D334 A/S 3073 3082 B/D
2317
T1
D< D78 K41
M13
M17
M17 A/S 2371 2378 2390 2393 2409 B/D 2368 2380 2397 2412 2416
M17 A/S 2453 2485 2517 2619 2669 B/D 2645 2704 2763 2859 2931
M17 A/S 2678 2728 2737 2787 2796 B/D 2966 3001 3093
M17 A/S 2926 2934 2961 2969 2996 B/D
M17 A/S 3004 3031 B/D
D> D340 K0
M51
M18
M18 A/S 2372 2381 2391 2398 2413 B/D 2369 2379 2394 2410 2417
M18 A/S 2456 2488 2520 2644 2670 B/D 2620 2679 2738 2797 2860
M18 A/S 2703 2729 2762 2788 2930 B/D 2927 2962 2997
M18 A/S 2935 2965 2970 3000 3005 B/D
D< D130 K90
M19
M19 A/S 2353 2373 2395 2399 2418 B/D 2392 2411 2414
M19 A/S 2459 2491 2523 2673 2732 B/D
M19 A/S 2791 B/D
2353
M19
D< D122 K70
M23
M24
M24 A/S 2377 2401 B/D 2375
M23 M17 M18
M25
M25 A/S 2374 B/D 2402
2371
M17 M18 M19 M11
Y004
M25 M24
M20
M20 A/S 2425 2549 2921 A/S
M24 M17 M18
M17 M18

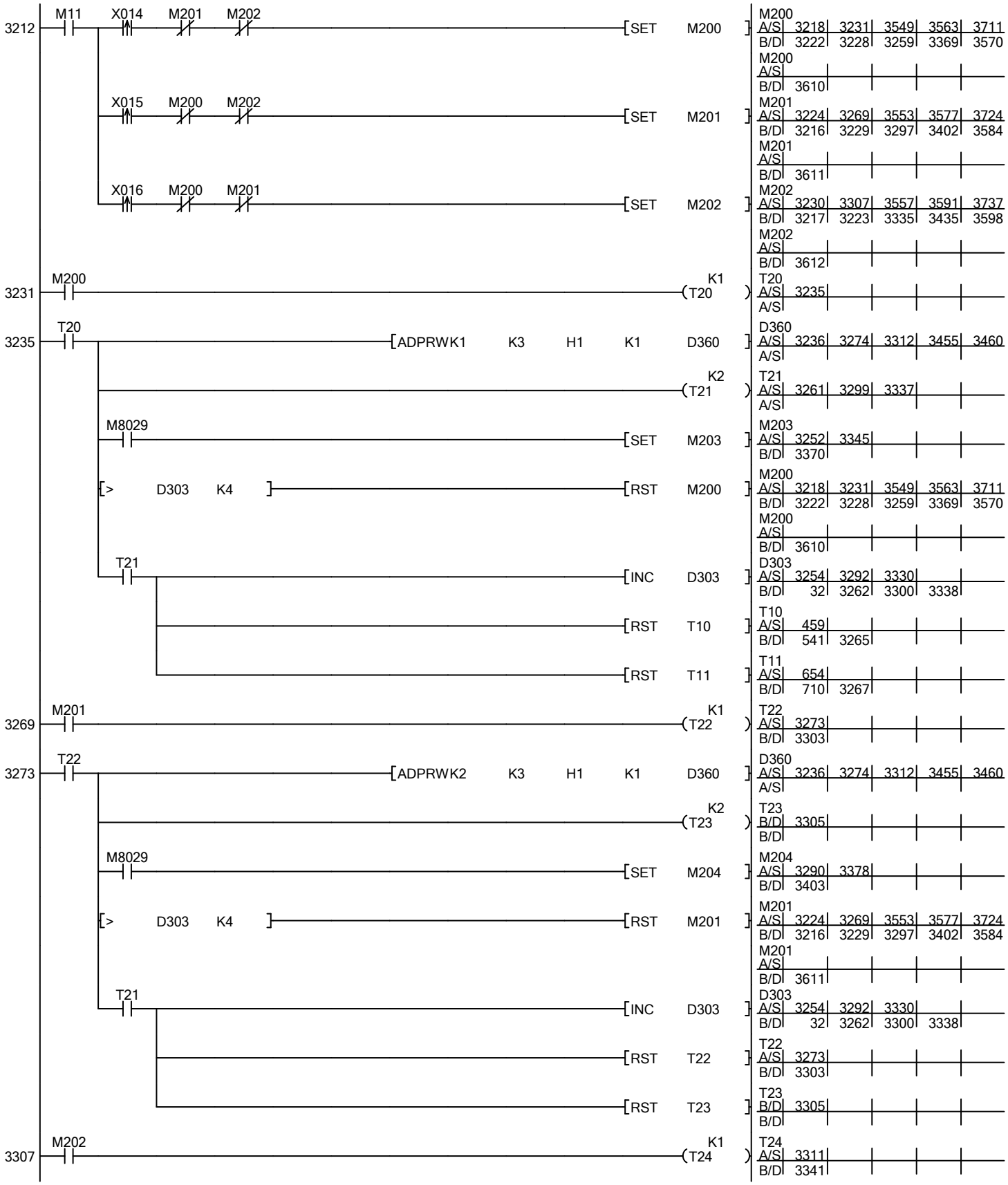

2390
M17 M18 M19 M11 Y003
M17 M18 M19 M21
M21 A/S 2433 2615 A/S
M17 M18 M19
M24 M25
2409
M17 M18 M19 M11 Y002
M17 M18 M19 M22
M22 A/S 2441 2449 2552 B/D 2794 3029
M17 M18 M19
2425
M20 [$MOV "FSM" D480]
D480 B/D 2426 2434 2442 B/D
2433
M21 [$MOV "HSM" D480]
D480 B/D 2426 2434 2442 B/D
2441
M22 [$MOV "LSM" D480]
D480 B/D 2426 2434 2442 B/D
2449
M22 M14 M17 M100
M100 A/S 3095 A/S
M18 M101
M101 A/S 3106 A/S
M19 [D> D332 D334] M102
M102 A/S 3117 A/S
[D> D334 D332] M103
M103 A/S 3128 A/S
M15 M17 M100
M100 A/S 3095 A/S
M18 M102
M102 A/S 3117 A/S
M19 [D> D330 D334] M101
M101 A/S 3106 A/S
[D> D334 D330] M103
M103 A/S 3128 A/S
M16 M17 M100
M100 A/S 3095 A/S
M18 M103
M103 A/S 3128 A/S
M19 [D> D330 D332] M101
M101 A/S 3106 A/S
[D> D332 D330] M102
M102 A/S 3117 A/S
M23 M20 M100
M100 A/S 3095 A/S
M22 M117
M117 A/S 2554 A/S

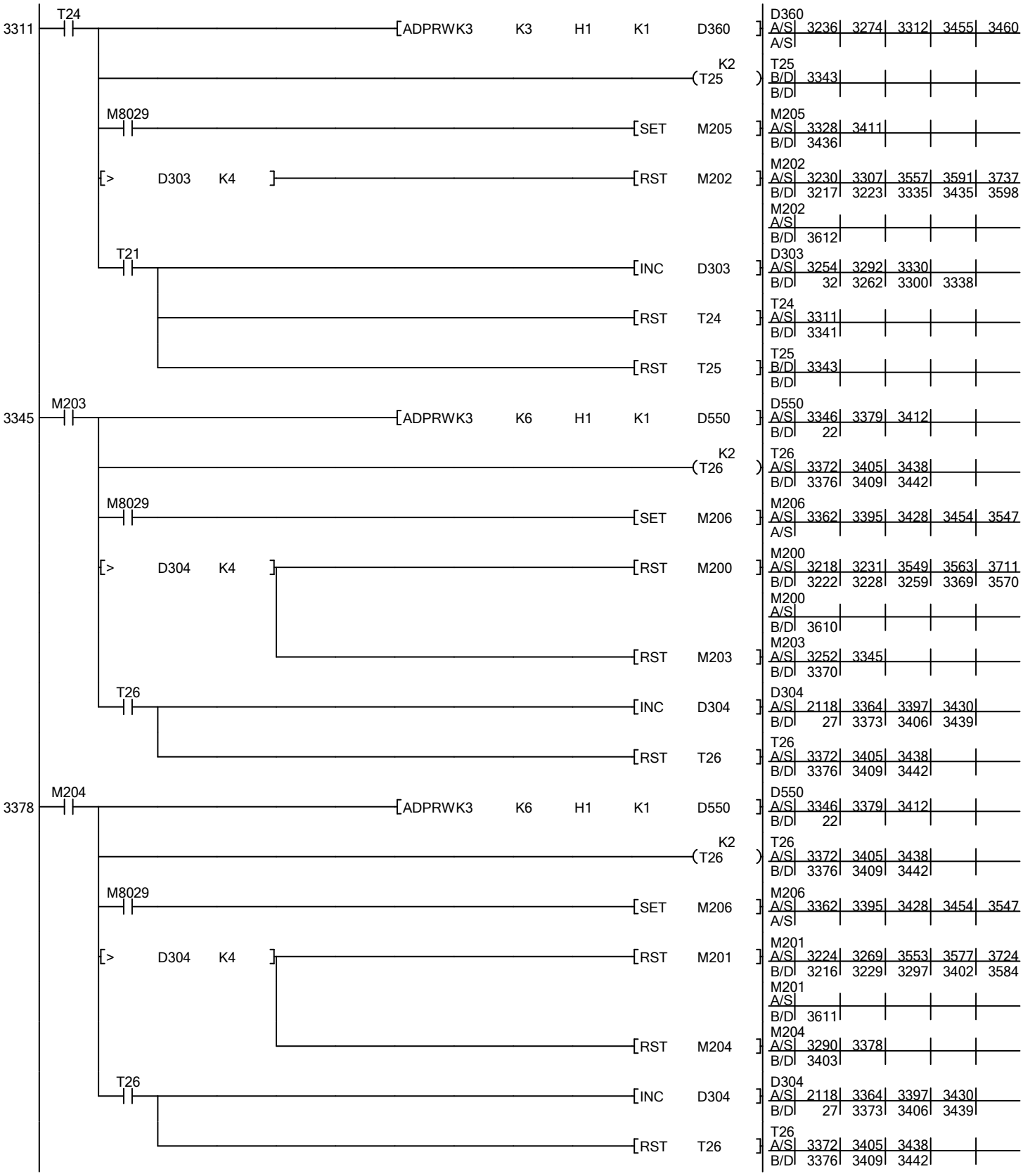

2554
M117
D>
D330
D332
D>
D330
D334
M101
D>
D332
D330
D>
D332
D334
M102
D>
D334
D330
D>
D334
D332
M103
M101
A/S
3106
A/S
M102
A/S
3117
A/S
M103
A/S
3128
A/S

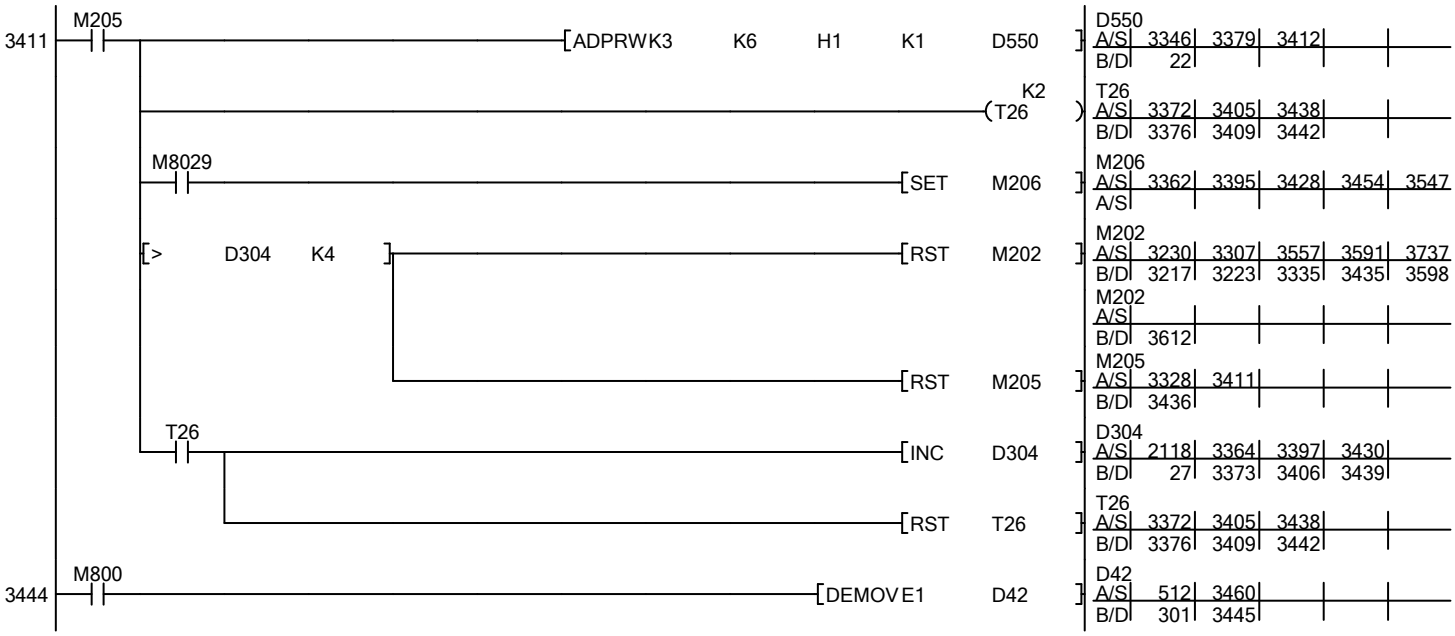

2615
M21 M14 M17 M18 D> D332 D334 M105
D> D334 D332 M106
M18 M17 D> D332 D334 M107
D> D334 D332 M108
M17 M18 M104
M19 M109
M15 M17 M18 D> D330 D334 M104
D> D334 D330 M106
M18 M17 D> D330 D334 M107
D> D334 D330 M109
M17 M18 M105
M19 M108
M16 M17 M18 D> D330 D332 M104
D> D332 D330 M105
M18 M17 D> D330 D332 M108
D> D332 D330 M109
M17 M18 M106
M19 M107
M22 M17 M18 D> D330 D332 D> D330 D334 M104
D> D332 D330 D> D332 D334 M105
D> D334 D330 D> D334 D332 M106
M17 M18 D< D330 D332 D< D330 D334 M109
D< D332 D330 D< D332 D334 M108
D< D334 D330 D< D334 D332 M107
M105 A/S 3147 A/S
M106 A/S 3155 A/S
M107 A/S 3163 A/S
M108 A/S 3171 A/S
M104 A/S 3139 A/S
M109 A/S 3179 A/S

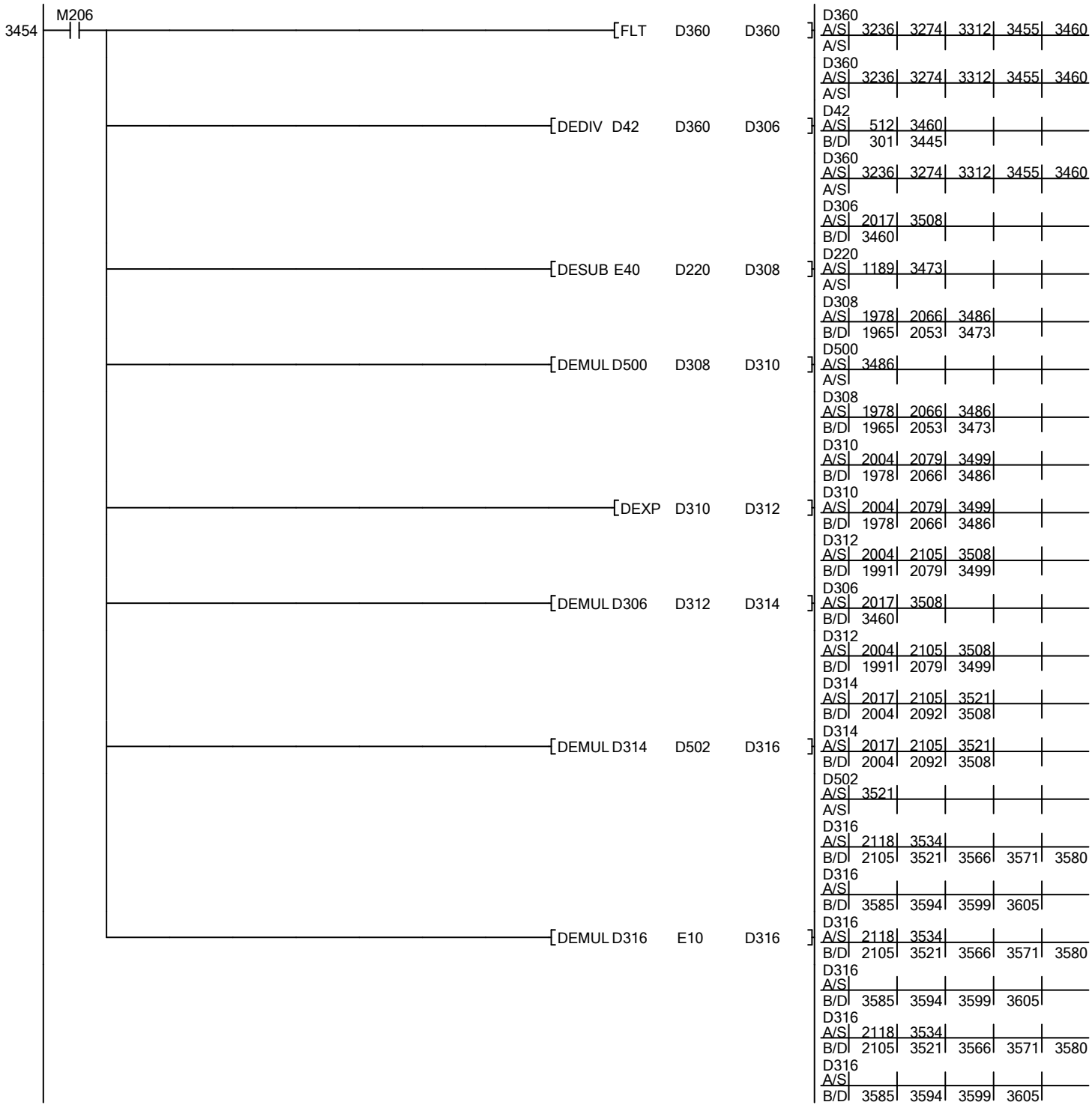

2921
M20 M11 M14 M17 M18 (Y012)
M18 M17 (Y011)
M17 M18 [D> D332 D334] (Y014)
[D> D334 D332] (Y013)
M15 M17 M18 (Y013)
M18 M17 (Y011)
M17 M18 [D> D330 D334] (Y014)
[D> D334 D330] (Y012)
M16 M17 M18 (Y014)
M18 M17 (Y011)
M17 M18 [D> D330 D332] (Y013)
[D> D332 D330] (Y012)
M22 M17 [D< D330 D332][D< D330 D334] (Y012)
[D< D332 D330][D< D332 D334] (Y013)
[D< D334 D330][D< D334 D332] (Y014)
M17 (Y011)
3095
M100 M11 (Y012)
(Y013)
(Y014)
3106
M101 M11 (Y011)
(Y013)
(Y014)

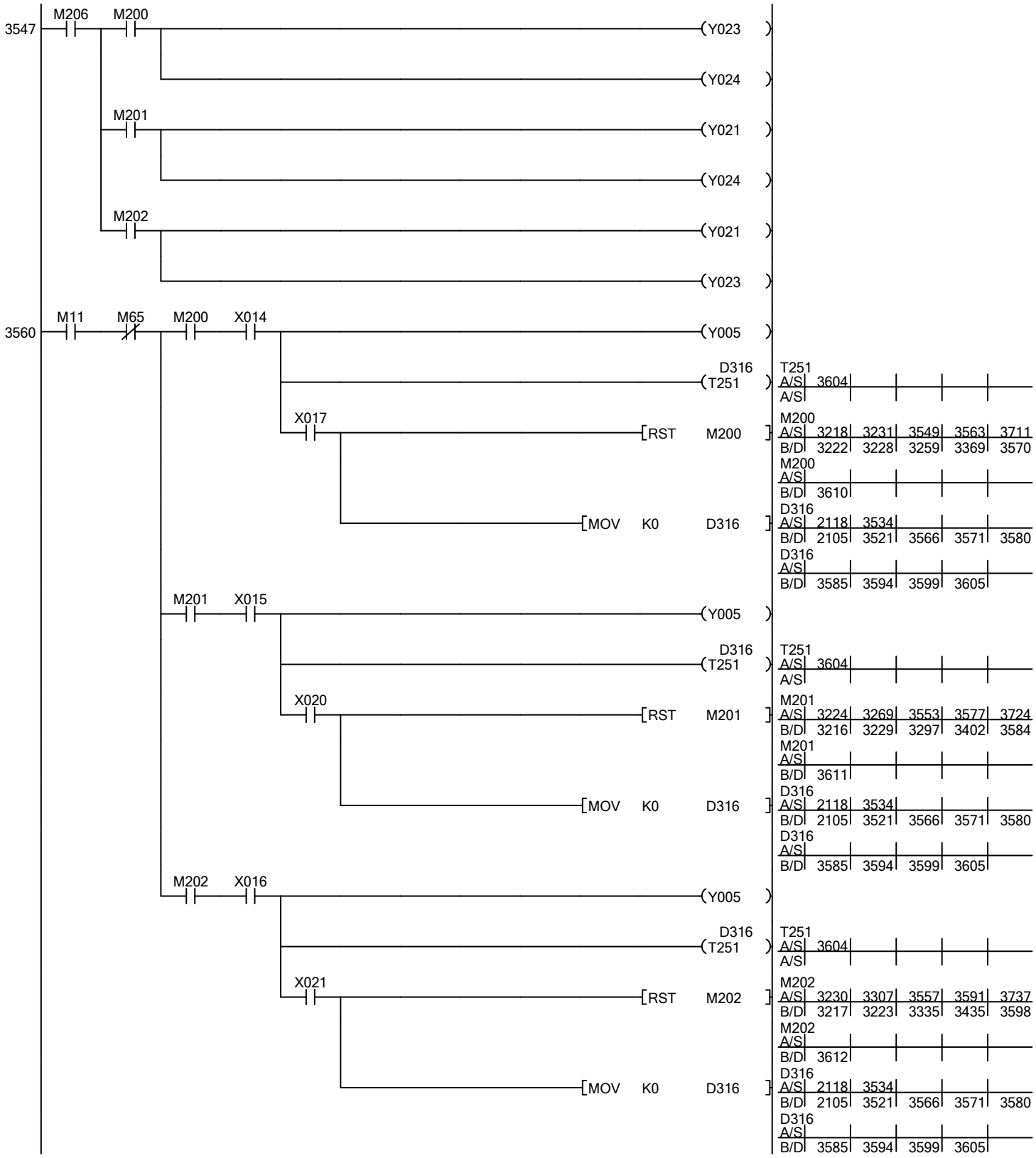

3117
M102
M11
Y011
Y012
Y014
3128
M103
M11
Y011
Y012
Y013
3139
M104
M11
Y013
Y014
3147
M105
M11
Y012
Y014
3155
M106
M11
Y012
Y013
3163
M107
M11
Y011
Y014
3171
M108
M11
Y011
Y013
3179
M109
M11
Y011
Y012
3187
M11
D< D178 D86
Y020
Y013
D> D178 D86
Y021
Y013

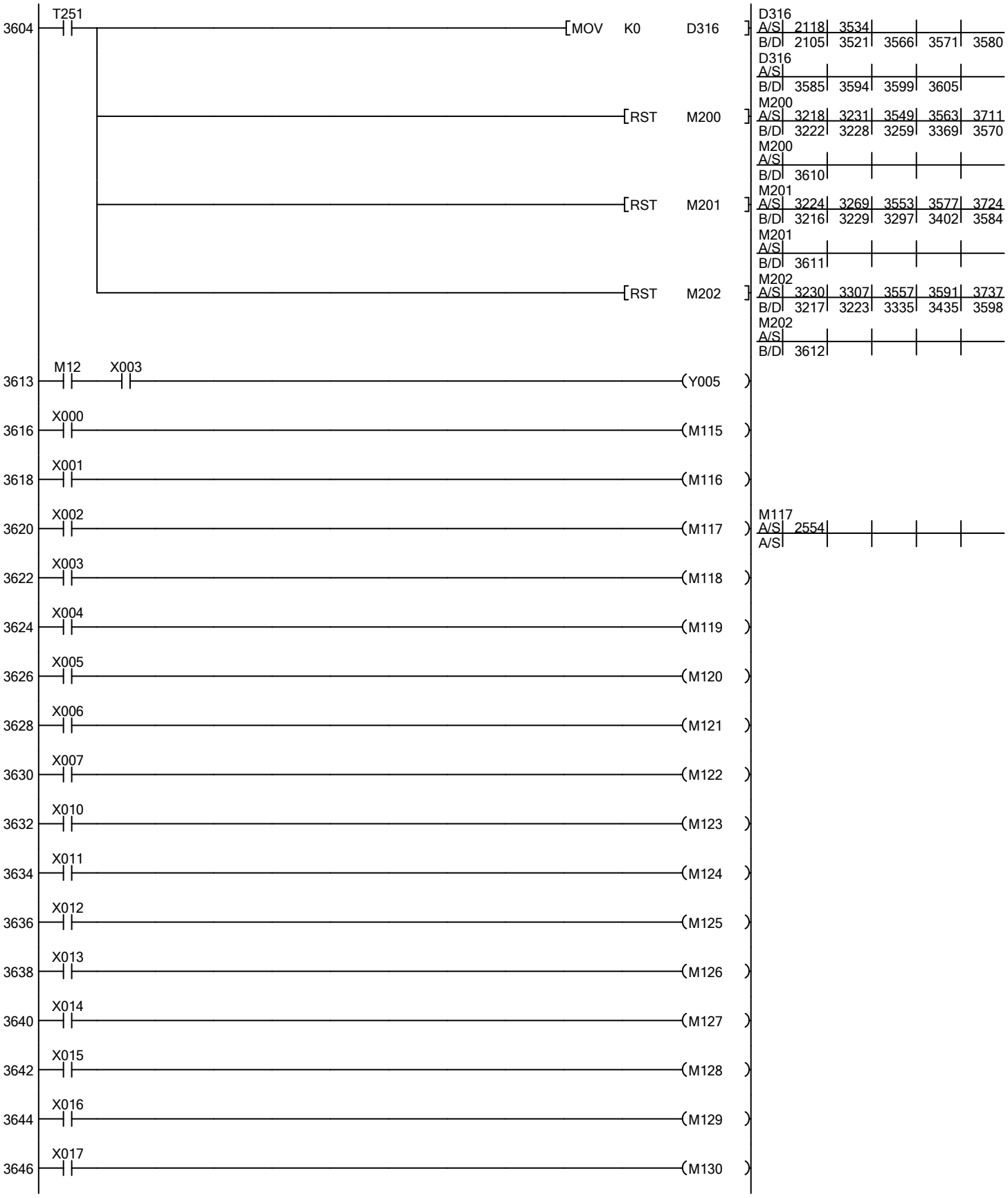

3212
M11 X014 M201 M202 SET M200
M200 A/S 3218 3231 3549 3563 3711 B/D 3222 3228 3259 3369 3570
M200 A/S B/D 3610
X015 M200 M202 SET M201
M201 A/S 3224 3269 3553 3577 3724 B/D 3216 3229 3297 3402 3584
M201 A/S B/D 3611
X016 M200 M201 SET M202
M202 A/S 3230 3307 3557 3591 3737 B/D 3217 3223 3335 3435 3598
M202 A/S B/D 3612
3231
M200 K1 T20
T20 A/S 3235 A/S
3235
T20 ADPRWK1 K3 H1 K1 D360
D360 A/S 3236 3274 3312 3455 3460 A/S
K2 T21
T21 A/S 3261 3299 3337 A/S
M8029 SET M203
M203 A/S 3252 3345 B/D 3370
> D303 K4 RST M200
M200 A/S 3218 3231 3549 3563 3711 B/D 3222 3228 3259 3369 3570
M200 A/S B/D 3610
T21 INC D303
D303 A/S 3254 3292 3330 B/D 32 3262 3300 3338
RST T10
T10 A/S 459 B/D 541 3265
RST T11
T11 A/S 654 B/D 710 3267
3269
M201 K1 T22
T22 A/S 3273 B/D 3303
3273
T22 ADPRWK2 K3 H1 K1 D360
D360 A/S 3236 3274 3312 3455 3460 A/S
K2 T23
T23 B/D 3305 B/D
M8029 SET M204
M204 A/S 3290 3378 B/D 3403
> D303 K4 RST M201
M201 A/S 3224 3269 3553 3577 3724 B/D 3216 3229 3297 3402 3584
M201 A/S B/D 3611
T21 INC D303
D303 A/S 3254 3292 3330 B/D 32 3262 3300 3338
RST T22
T22 A/S 3273 B/D 3303
RST T23
T23 B/D 3305 B/D
3307
M202 K1 T24
T24 A/S 3311 B/D 3341

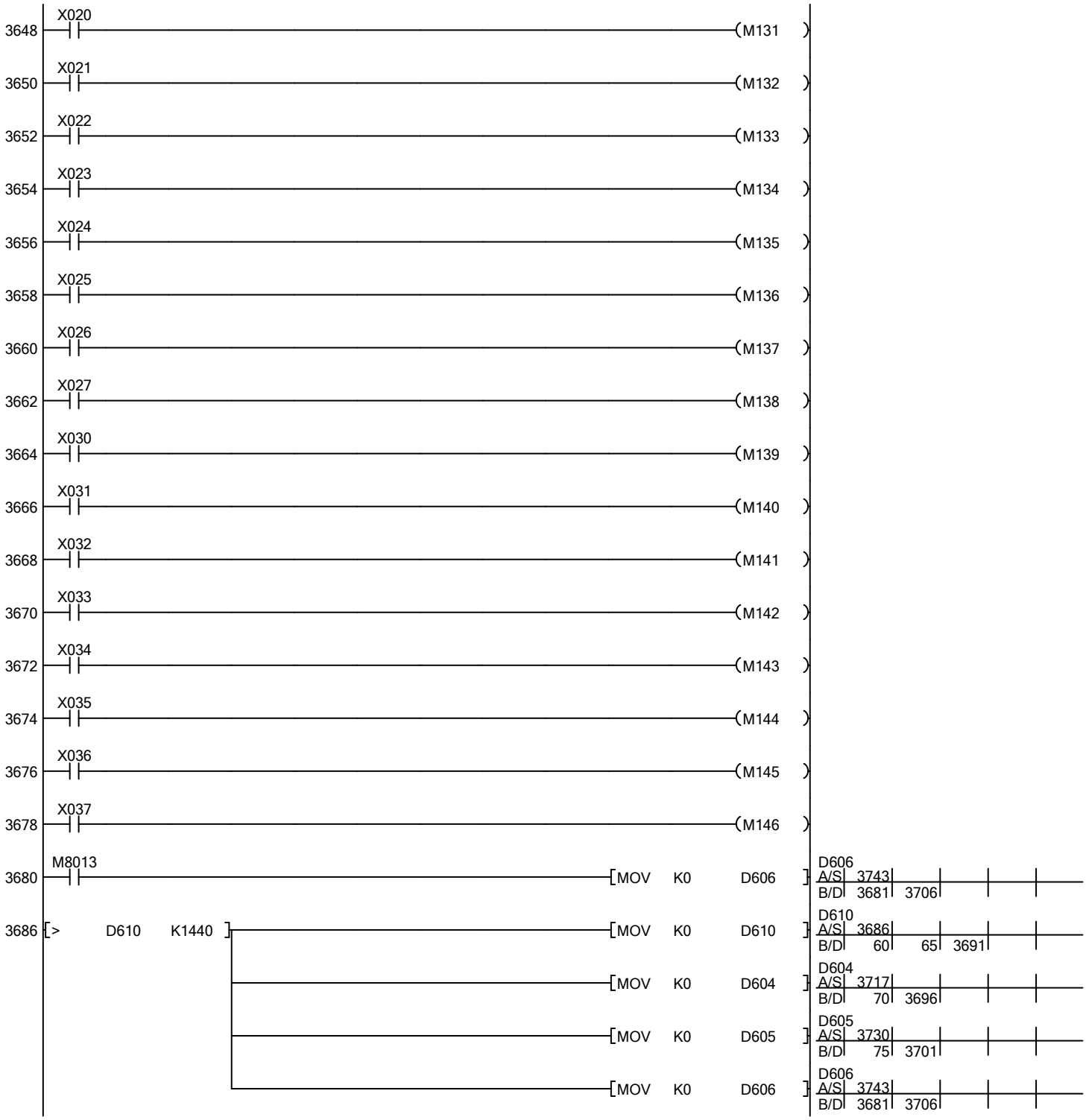

3311
T24
ADPRW K3 K3 H1 K1 D360
K2
T25
M8029
SET M205
> D303 K4
RST M202
T21
INC D303
RST T24
RST T25
3345
M203
ADPRW K3 K6 H1 K1 D550
K2
T26
M8029
SET M206
> D304 K4
RST M200
RST M203
T26
INC D304
RST T26
3378
M204
ADPRW K3 K6 H1 K1 D550
K2
T26
M8029
SET M206
> D304 K4
RST M201
RST M204
T26
INC D304
RST T26
D360 A/S 3236 3274 3312 3455 3460 A/S
T25 B/D 3343 B/D
M205 A/S 3328 3411 B/D 3436
M202 A/S 3230 3307 3557 3591 3737 B/D 3217 3223 3335 3435 3598
M202 A/S B/D 3612
D303 A/S 3254 3292 3330 B/D 32 3262 3300 3338
T24 A/S 3311 B/D 3341
T25 B/D 3343 B/D
D550 A/S 3346 3379 3412 B/D 22
T26 A/S 3372 3405 3438 B/D 3376 3409 3442
M206 A/S 3362 3395 3428 3454 3547 A/S
M200 A/S 3218 3231 3549 3563 3711 B/D 3222 3228 3259 3369 3570
M200 A/S B/D 3610
M203 A/S 3252 3345 B/D 3370
D304 A/S 2118 3364 3397 3430 B/D 27 3373 3406 3439
T26 A/S 3372 3405 3438 B/D 3376 3409 3442
D550 A/S 3346 3379 3412 B/D 22
T26 A/S 3372 3405 3438 B/D 3376 3409 3442
M206 A/S 3362 3395 3428 3454 3547 A/S
M201 A/S 3224 3269 3553 3577 3724 B/D 3216 3229 3297 3402 3584
M201 A/S B/D 3611
M204 A/S 3290 3378 B/D 3403
D304 A/S 2118 3364 3397 3430 B/D 27 3373 3406 3439
T26 A/S 3372 3405 3438 B/D 3376 3409 3442

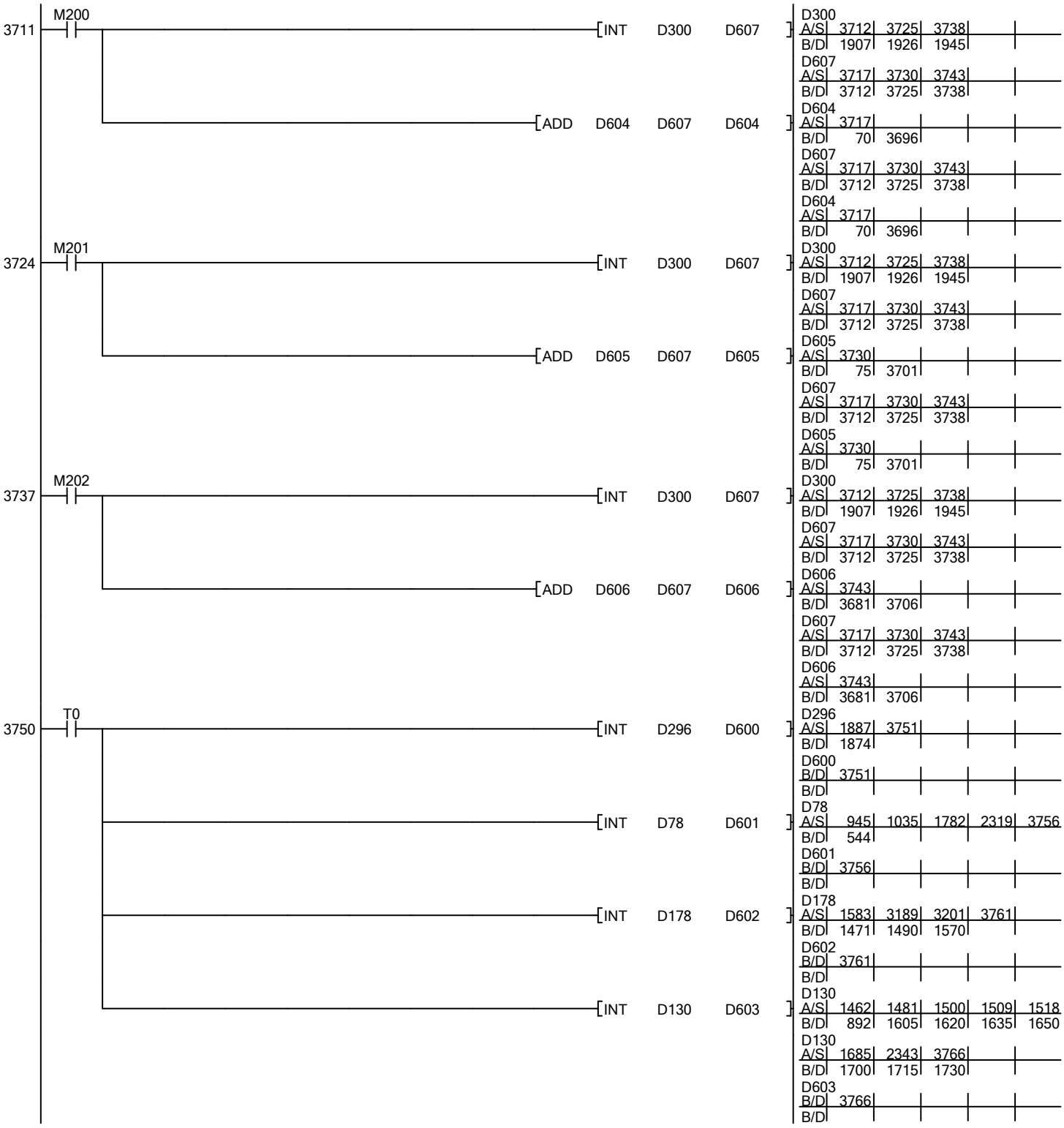

3411
M205
ADPRW K3 K6 H1 K1 D550
K2
T26
M8029
SET M206
> D304 K4
RST M202
RST M205
T26
INC D304
RST T26
3444
M800
DEMOV E1 D42
D550 A/S 3346 3379 3412 B/D 22
T26 A/S 3372 3405 3438 B/D 3376 3409 3442
M206 A/S 3362 3395 3428 3454 3547 A/S
M202 A/S 3230 3307 3557 3591 3737 B/D 3217 3223 3335 3435 3598
M202 A/S B/D 3612
M205 A/S 3328 3411 B/D 3436
D304 A/S 2118 3364 3397 3430 B/D 27 3373 3406 3439
T26 A/S 3372 3405 3438 B/D 3376 3409 3442
D42 A/S 512 3460 B/D 301 3445

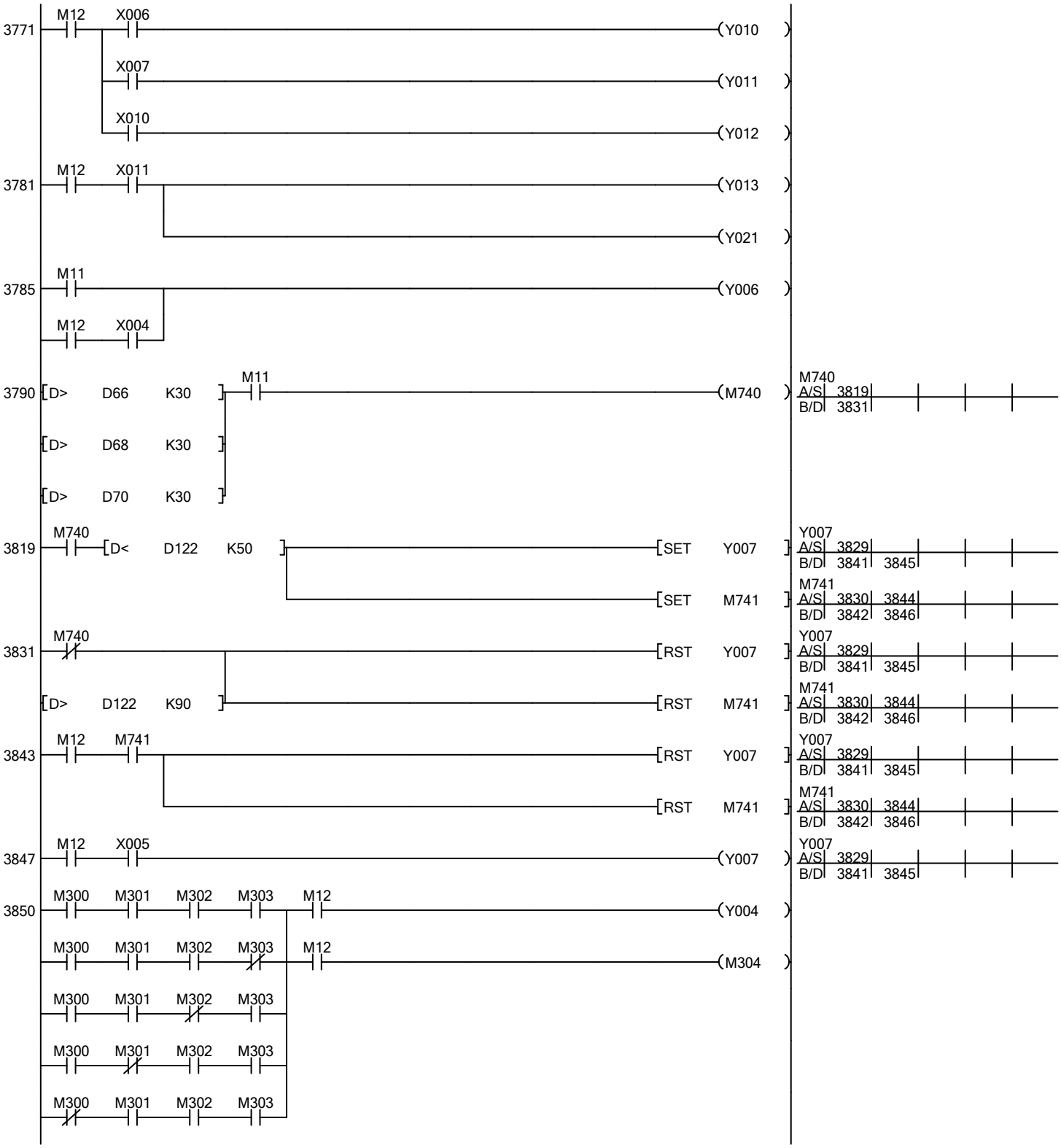

3454
M206
FLT D360 D360
D360
A/S 3236 3274 3312 3455 3460
A/S
D360
A/S 3236 3274 3312 3455 3460
A/S
DEDIV D42 D360 D306
D42
A/S 512 3460
B/D 301 3445
D360
A/S 3236 3274 3312 3455 3460
A/S
D306
A/S 2017 3508
B/D 3460
DESUB E40 D220 D308
D220
A/S 1189 3473
A/S
D308
A/S 1978 2066 3486
B/D 1965 2053 3473
DEMUL D500 D308 D310
D500
A/S 3486
A/S
D308
A/S 1978 2066 3486
B/D 1965 2053 3473
D310
A/S 2004 2079 3499
B/D 1978 2066 3486
DEXP D310 D312
D310
A/S 2004 2079 3499
B/D 1978 2066 3486
D312
A/S 2004 2105 3508
B/D 1991 2079 3499
DEMUL D306 D312 D314
D306
A/S 2017 3508
B/D 3460
D312
A/S 2004 2105 3508
B/D 1991 2079 3499
D314
A/S 2017 2105 3521
B/D 2004 2092 3508
DEMUL D314 D502 D316
D314
A/S 2017 2105 3521
B/D 2004 2092 3508
D502
A/S 3521
A/S
D316
A/S 2118 3534
B/D 2105 3521 3566 3571 3580
D316
A/S
B/D 3585 3594 3599 3605
DEMUL D316 E10 D316
D316
A/S 2118 3534
B/D 2105 3521 3566 3571 3580
D316
A/S
B/D 3585 3594 3599 3605
D316
A/S 2118 3534
B/D 2105 3521 3566 3571 3580
D316
A/S
B/D 3585 3594 3599 3605

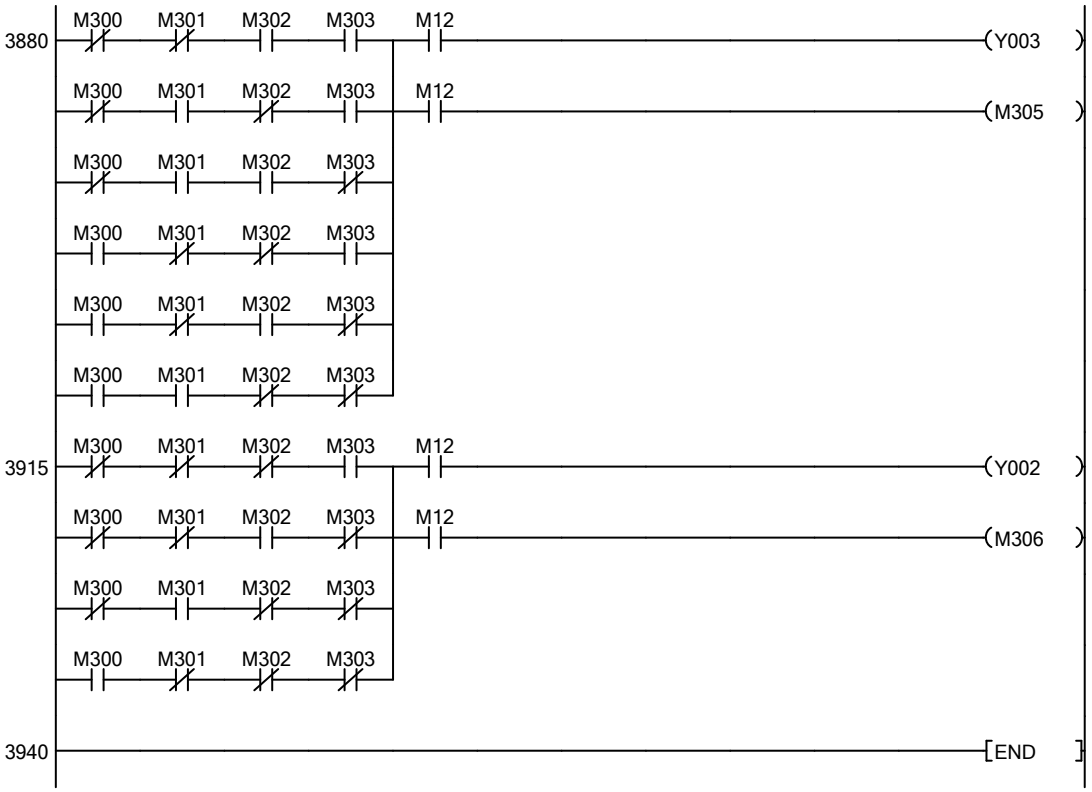

3547
M206
M200
Y023
Y024
M201
Y021
Y024
M202
Y021
Y023
3560
M11
M65
M200
X014
Y005
D316
T251
T251
A/S 3604
A/S
X017
RST M200
M200
A/S 3218 3231 3549 3563 3711
B/D 3222 3228 3259 3369 3570
M200
A/S
B/D 3610
MOV K0 D316
D316
A/S 2118 3534
B/D 2105 3521 3566 3571 3580
D316
A/S
B/D 3585 3594 3599 3605
M201
X015
Y005
D316
T251
T251
A/S 3604
A/S
X020
RST M201
M201
A/S 3224 3269 3553 3577 3724
B/D 3216 3229 3297 3402 3584
M201
A/S
B/D 3611
MOV K0 D316
D316
A/S 2118 3534
B/D 2105 3521 3566 3571 3580
D316
A/S
B/D 3585 3594 3599 3605
M202
X016
Y005
D316
T251
T251
A/S 3604
A/S
X021
RST M202
M202
A/S 3230 3307 3557 3591 3737
B/D 3217 3223 3335 3435 3598
M202
A/S
B/D 3612
MOV K0 D316
D316
A/S 2118 3534
B/D 2105 3521 3566 3571 3580
D316
A/S
B/D 3585 3594 3599 3605


3604 T251 [MOV K0 D316]
[RST M200]
[RST M201]
[RST M202]
D316 A/S 2118 3534 B/D 2105 3521 3566 3571 3580
D316 A/S B/D 3585 3594 3599 3605
M200 A/S 3218 3231 3549 3563 3711 B/D 3222 3228 3259 3369 3570
M200 A/S B/D 3610
M201 A/S 3224 3269 3553 3577 3724 B/D 3216 3229 3297 3402 3584
M201 A/S B/D 3611
M202 A/S 3230 3307 3557 3591 3737 B/D 3217 3223 3335 3435 3598
M202 A/S B/D 3612
3613 M12 X003 (Y005)
3616 X000 (M115)
3618 X001 (M116)
3620 X002 (M117)
M117 A/S 2554 A/S
3622 X003 (M118)
3624 X004 (M119)
3626 X005 (M120)
3628 X006 (M121)
3630 X007 (M122)
3632 X010 (M123)
3634 X011 (M124)
3636 X012 (M125)
3638 X013 (M126)
3640 X014 (M127)
3642 X015 (M128)
3644 X016 (M129)
3646 X017 (M130)


3648 X020 M131
3650 X021 M132
3652 X022 M133
3654 X023 M134
3656 X024 M135
3658 X025 M136
3660 X026 M137
3662 X027 M138
3664 X030 M139
3666 X031 M140
3668 X032 M141
3670 X033 M142
3672 X034 M143
3674 X035 M144
3676 X036 M145
3678 X037 M146
3680 M8013 [MOV K0 D606]
3686 [> D610 K1440]
[MOV K0 D610]
[MOV K0 D604]
[MOV K0 D605]
[MOV K0 D606]
D606 A/S 3743 B/D 3681 3706
D610 A/S 3686 B/D 60 65 3691
D604 A/S 3717 B/D 70 3696
D605 A/S 3730 B/D 75 3701
D606 A/S 3743 B/D 3681 3706


3711
M200
INT D300 D607
D300 A/S 3712 3725 3738 B/D 1907 1926 1945
D607 A/S 3717 3730 3743 B/D 3712 3725 3738
ADD D604 D607 D604
D604 A/S 3717 B/D 70 3696
D607 A/S 3717 3730 3743 B/D 3712 3725 3738
D604 A/S 3717 B/D 70 3696
3724
M201
INT D300 D607
D300 A/S 3712 3725 3738 B/D 1907 1926 1945
D607 A/S 3717 3730 3743 B/D 3712 3725 3738
ADD D605 D607 D605
D605 A/S 3730 B/D 75 3701
D607 A/S 3717 3730 3743 B/D 3712 3725 3738
D605 A/S 3730 B/D 75 3701
3737
M202
INT D300 D607
D300 A/S 3712 3725 3738 B/D 1907 1926 1945
D607 A/S 3717 3730 3743 B/D 3712 3725 3738
ADD D606 D607 D606
D606 A/S 3743 B/D 3681 3706
D607 A/S 3717 3730 3743 B/D 3712 3725 3738
D606 A/S 3743 B/D 3681 3706
3750
T0
INT D296 D600
D296 A/S 1887 3751 B/D 1874
D600 B/D 3751 B/D
INT D78 D601
D78 A/S 945 1035 1782 2319 3756 B/D 544
D601 B/D 3756 B/D
INT D178 D602
D178 A/S 1583 3189 3201 3761 B/D 1471 1490 1570
D602 B/D 3761 B/D
INT D130 D603
D130 A/S 1462 1481 1500 1509 1518 B/D 892 1605 1620 1635 1650
D130 A/S 1685 2343 3766 B/D 1700 1715 1730
D603 B/D 3766 B/D


3771
M12 X006 Y010
X007 Y011
X010 Y012
3781
M12 X011 Y013
Y021
3785
M11 Y006
M12 X004
3790
D> D66 K30
D> D68 K30
D> D70 K30
M11 M740
M740 A/S 3819 B/D 3831
3819
M740 D< D122 K50
SET Y007
SET M741
Y007 A/S 3829 B/D 3841 3845
M741 A/S 3830 3844 B/D 3842 3846
3831
M740
RST Y007
D> D122 K90
RST M741
Y007 A/S 3829 B/D 3841 3845
M741 A/S 3830 3844 B/D 3842 3846
3843
M12 M741
RST Y007
RST M741
Y007 A/S 3829 B/D 3841 3845
M741 A/S 3830 3844 B/D 3842 3846
3847
M12 X005 Y007
Y007 A/S 3829 B/D 3841 3845
3850
M300 M301 M302 M303 M12 Y004
M300 M301 M302 M303 M12 M304
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303


3880
M300 M301 M302 M303 M12 (Y003)
M300 M301 M302 M303 M12 (M305)
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
3915
M300 M301 M302 M303 M12 (Y002)
M300 M301 M302 M303 M12 (M306)
M300 M301 M302 M303
M300 M301 M302 M303
3940
[END]

[Line Statement List]

Step	Line Statement
0	Timer
16	Declarations
80	Analog to Digital
114	ADC to Float
170	Temperature
327	Distance
432	Flowmeter
459	Temperature Averaging
654	Level Averaging
944	5 - sec Sensor Logging
1167	Error Threshold
1438	E-Stops
1451	Automation
1460	Water Temperature logging
1593	Heater latching
1768	Height to Mass Conversion
1821	Dynamic Upper Thermal Boundary
1897	PO Desingation
1954	PO Timer
2144	PO Dispensing
2162	Tank scoring
2317	Need Scheduler
2371	Mode Scheduler
2425	Mode Indicator
2449	LSM Scheduler
2615	HSM Scheduler
2921	FSM Scheduler
3095	Scheduler Gates
3187	Solar gates
3212	Oil Scheduler
3444	Oil Calculation
3547	Oil Distribution Gating
3560	Oil Distribution Timing
3616	HMI Display for Inputs
3680	Oil Chart
3750	Charting
3771	Manual Heater Control
3781	Manual Solar Gate
3785	Jacket pump (Auto/manual)
3790	AODD (Auto/Manual)
3850	Manual Pump Mode